

# Hello Apathy

*Nicolas Jao*

## Part 1

I want to make sure you understand: this story's a warning, so please listen carefully.

“Dr. Wang, please hand me the USB drive.”

“I didn't graduate with the Apex Scholar award at one of the top schools in the nation just to become your teacher at this company, Josie.”

Dr. Yao laughed. “Listen, we get it. We know you miss your intellectual soulmate because she transferred to another school in your final year, but you don't have to ramble on about it to our faces. General Zhao approved of this plan, and you don't want to get on his bad side. Believe me, I've experienced it firsthand. Now, hand me the drive so we can do our job, get out of here as fast as possible, and go back to working on our projects.”

There was a time when, if I got sick, you and your mother would visit me in bed and bring me soup. It was a bold and deliberate move, the journey sometimes cold or against a thunderous storm or relentless rain. But you made it every time, because in those days we lived close to each other, and from all the homework, after-school supplementary classes, piano, flute, swimming, and Wushu, we probably only had each other to talk to. The other kids at school had different, rigid schedules that seldom aligned with ours. Although I didn't think it at the time, it was only natural that we became friends; it's only reasonable to assume it based on the abundance of exposure to each other and our families.

“Oh, would you shut up about that. I told you already, that wasn't the reason I was upset. It wasn't even about losing a rival. At first, I was happy to be the only one left unchallenged in the class; I even won bets off my friends because of it. But perhaps a few weeks later, I realized the inequity of it all. There was an imbalance she left once she was gone, and

I hated that this was her legacy, that she went down without much of a fight due to her family's circumstances. There is not much honour in my victory, so I don't blame her at all."

"Do you know how she is doing now?"

"Let's refocus on our task, Dr. Yao. Look at you, suddenly interested in my life."

Dr. Yao frowned and plugged the USB drive into the computer. We waited for it to load. Once it did, she got to work typing a program with the keyboard. Dr. Wang and I sat patiently behind her chair, our hands in the pockets of our white lab coats, watching her create code like a composer weaving notes together on a sheet of paper, bringing music to life in the form of an executable compilation of binary 1's and 0's. She spoke to the machine in silence, creating commands for it born out of the complexities of her mind, and I observed every detail. I was still in training.

I made it a big deal that I got better grades than you, and my mother was very proud of that. She talked with your mother a lot, and they compared us all the time, as mothers do, and I always felt bad for you because you never gave your mother much to work with to put up a fight against mine. Sometimes I would overhear their conversation, my mother continuously humble-bragging about my achievements, and I would chime in and say that you caught a frog today, something nobody else in our class could do, or that you sketched a really beautiful drawing of a lady from your favourite show in your notebook, or even that you climbed to a rank higher than anyone else I knew in our favourite online RPG. But all your mother would say was, "Why is he drawing in his notebook? That should be for class. If he wants to draw art, he should tell me to buy him a sketchbook."

I remember I would get angry whenever this happened. Angry at her, but also angry at my mother, I think. But of course, I never told you that. I would never admit it in a million years.

"Who knew I would get into researching particle physics just to get pulled aside for random side missions assigned by the government?" said Dr. Yao.

Dr. Wang retorted, "Particle physics is a dying field. There's nothing left to discover."

"That's not true! There's plenty of new particles out there that we don't know about; they just might be hard to discover. For example, there's a possibility that an unknown particle is what causes dark matter's effects on the universe, which could be explained by WIMPs or sterile neutrinos. There are leptiquarks—oh, and don't forget gravitons; that's an exciting one which could relate Einstein's gravity to the Standard Model. We still have the strong CP problem in quantum chromodynamics, and axions were proposed to be a possible solution. All these particles—although they haven't been detected, it's possible there are many good ones out there, but I've still yet to find them."

"Is that what you say about your failed love life as well?"

"Jin, I'm so close to wrapping this wired mouse around your neck and stabbing your

throat with this USB stick so you can use that big mouth of yours to spout out the code yourself.”

Dr. Wang laughed.

Back then you didn't seem like a smart kid. I was focused on those high-stakes regional and national tests to make it to my dream high school and university, but you were a rowdy type of boy who never seemed to care about classes as much as I did. If nothing else, the way I knew you, you were out for adventure. You always insisted I come with you to sneak into the mansion at the top of that hill in our village because the family there had a better piano than yours or mine. The first time we met was because of a piano. We had the same piano teacher, and you were the student before me. During one of your practices, my mother brought me there early, and I listened to you play beautifully from outside the door. While my mother chatted with our piano teacher, I couldn't help but sneak into the room and watch you play. I don't remember who said hello first. But I must have stared at you, without you knowing you were being watched, for a few minutes before one of us did. Then we became friends once we learned we went to the same school.

The mansion we always snuck into rested at the top of a hill on the outskirts of our rural village. To get there, we had to bike past dirty streets with heavy power lines and swamps with wooden boardwalks lit by fireflies at night. We had to use the rest of our energy cycling uphill to reach the mansion, and when we looked back, we could see the lights of our entire town below. The interesting part about this mansion was that the owners were never home. They were rich and had multiple homes, and quite oddly, there was no real protection on this one. If there was a housekeeper there, they ignored us. They didn't particularly care about two kids sneaking in through an unlocked window. I've always thought perhaps they stole things from the owners as well.

“Dr. Chen, can you track Dr. Lei for me? She's delivering the DC power supply and the cracked version of Commshark, but she was supposed to arrive half an hour ago,” said Dr. Yao.

“Oh.” She was talking to me. “Track her? How do I do that?”

“It's part of this company-wide system Lumina uses to ensure ‘maximum productivity.’” She rolled her eyes. “Boss implemented it. It's called TeamSync. We use it all the time to track each other's locations. There's a link to access it on our intranet.”

“OK.”

“We didn't bring the encrypted briefcase laptop with Commshark because it had to go to the administrative office to get the passcode authorized first,” said Dr. Wang. “As for the extra power supply to help our model over here run its parallel computations faster...” He put a hand to the back of his head. “We simply forgot that. As a result, we have her running both errands.”

Dr. Yao glared at him. “We could have started the deep learning process right now if you didn’t forget it, Jin. You said we needed double the power to run this model, and I reminded you last night!”

He laughed. “I stayed up gathering enough resources for our base so we can take on that high-level monster tomorrow. At some point, today’s task slipped my mind.”

Dr. Yao stood up from her chair, turned, and whispered to me, “I’m sure he forgot about it until General Zhao visited the office this morning. That’s why he spilled his tea over his cable-knit pullover from his alma mater at the sudden sight of him when we were in the meeting room.” I politely laughed. “Regardless, we need Dr. Lei to bring the items now. What’s taking her so long?”

“Josie, you haven’t been doing your raids lately, by the way.”

The mansion had a grand piano and a bar with expensive wines. What we’d do every time was spend time on the piano first, where we’d teach each other our favourite pieces, and then we’d get bored and start tasting the collection of wines at the bar. I admit it was your playing that drew me further into piano, but it was disheartening to see how, as time passed, I surpassed your skills, and you were content with mediocrity because you lost interest in practicing. I couldn’t blame you, given how many commitments our parents put both of us through. Regardless, we had fun, didn’t we? It was a cycle between reading textbooks at our school’s library, our numerous extracurricular activities, more often than not a big dinner with both our families where you’d take over control of the Lazy Susan to your father’s disapproval, and then with the little free time we’d have on weekends we’d bike to this mansion. We’d taste the wines at the bar one by one, drinking straight from the bottle, and laugh about how horrible each tasted. They were truly nasty, weren’t they? Then we’d carefully screw the caps back on and return them where we found them. We never touched the ones with corks because we didn’t want to figure out the corkscrew, and we weren’t sure if we could put them back on.

I remember one regretful night we didn’t end up reaching the mansion. On our way, while we were in the swampy area, we saw some teenagers, a boy and girl kissing on a bench facing a pond on the trail. We hid in some bushes to stare at them, and they were never aware of us. We saw them all over each other, and we laughed, thinking it was gross. But while I had left it at that, I believe it disturbed you. You treated it like an unacceptable, subpar experience. As if this sight was something we should not have seen, and additionally, something you never wanted to see again. The rest of the night left you in an irritated mood. You began to act rudely and annoyingly, then you brought up the topic of your lousy experiences with girls at school you liked, then we argued, and at its peak we decided it was best if we both just went home.

“TeamSync says she’ll be here in two minutes,” I said.

“Not bad,” said Dr. Wang. “Dr. Yao, finish up that neural network. It’s sloppy.”

“Oh, please. Are you saying you’re a bad teacher?”

“I’ve given you enough material to infer the rest. You’re a fast learner. Seriously, weren’t you beating 18-year-olds at math competitions when you were 8?”

Dr. Lei arrived. She rushed through the door, muttering something I barely heard: “I’m here!” Her voice was so soft she sounded like a mouse. Seeing her at the office was rare, so this was the first time I noticed. Dr. Yao told me she was shy, but this was when I witnessed that truth.

We winced at her clumsiness in handling the objects she carried. She placed the briefcase laptop and the power supply onto the table, almost dropping it. “Sorry I’m late! I had to explain to the general what I was here for. He had never met me...”

“Unbelievable,” said Dr. Yao, glaring at Dr. Wang. “You didn’t tell Zhao?”

Dr. Wang smirked, cautiously sliding the power supply off Dr. Lei’s hands on the table as if she were fragile. “Thanks.” Then he resumed his conversation with Dr. Yao: “You know, if our grant application fails for that new research project we plan to do, we could always start a dungeon-raiding service. I calculated our DPS last night—if we optimize our cooldown rotations, we could clear that boss in 2.7 minutes.”

“I’m going back to work now,” said Dr. Lei quietly. I was the only one who heard her. I watched her leave the room.

“Jin, you’d get distracted by the minion chests at the chance any of them could have good gear before that happens.”

“But if we applied machine learning to predict loot drops...”

It was after primary school that I discovered you were truly as smart as me. Sometimes, although I would never admit it to you, you were even smarter. With all my hard work, I made it to my dream high school, but the biggest shock was when my parents and I discovered you were going there too, and you didn’t tell me. Maybe you were scared to? Maybe you were trying to hide that you were as intelligent as me? I don’t know. I’ll be honest, it was probably only my dream high school because it was prestigious and my mother said it was a good school, and that I should try to go there. There was nothing else about it that made it stand out particularly to me, so when I learned that you were going there too, that all changed.

None of it mattered. Your grandfather got sick, and your father decided it was best for your whole family to move back to your home province to take care of him. We didn’t even get a chance to say goodbye. Family circumstances drove us apart, and I never saw you again.

Something Dr. Wang said made Dr. Yao laugh spontaneously.

“In 1752, Benjamin Franklin was famously credited with discovering the practical uses of electricity with his famous kite experiment,” said Dr. Yao, catching her breath. “Then American entrepreneurs were responsible for facilitating the country’s power grid and transmission lines, and the rest is history. If we discover a new particle with applications that

engineers and businessmen can commercialize to the world, we could have another similar revolution in our capabilities as a race. I'm talking bosons, quantum mechanics and general relativity, a theory of everything, new physics that hasn't been discovered or dreamt of yet. What if there are more than four fundamental forces past the ones in the Standard Model and gravity?"

"Is that so?" said Dr. Wang sarcastically, smirking. "How interesting. You speculate there's a fifth fundamental force in the universe? What could it be? Do you think..." He made an obscene gesture with his hands.

"Don't ridicule me, Jin," said Dr. Yao. The absurdity of the idea took a moment to register in her head, and when it did, she began to giggle. Then she tried to stop because it was Dr. Wang making her laugh, but she couldn't help herself. Dr. Wang began to laugh uncontrollably too. Soon enough, I joined them.

The rest of the night passed with no further issues. Dr. Yao was able to finish training the model to analyze the database's weaknesses. Dr. Wang told us to give him 30 minutes, and in that time, using what the model found, he wrote the code for a deadly virus, then released it into the enemy server. We saw General Zhao to verify our task had been completed, and he authorized the military to escort us back to Lumina's headquarters. We told them to take us to our favourite noodle place first. Once we grabbed the takeout, we returned to our office. Dr. Wang and Dr. Yao went downstairs to the labs to finish a project they were close to completing and would try to stay up until dawn, but I was too exhausted to watch them work. I went straight to my dorm, collapsed onto my bed, still in my lab coat and safety boots, and fell asleep.

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My name is Lily Chen, and I'm not having a good week. I'm incredibly tired. I'm using an old personal laptop I've kept in my childhood room's drawer to write this. It's a device I haven't used since going to university. At the speed it's going, it has about as much RAM as the toaster downstairs. But I suppose I have no choice, because my mother never threw it away, and I'm definitely not going to use my work computer for this. Everything you've read so far is what I've written. I'm telling this story at a point in time when I have already taken the cure. As such, it is important to understand this may not be the most accurate story from my telling, as it is contingent upon the complications of my memory. Stories are always slaves to memory. I've spent years with my coworker Dr. Lin studying the brain, so you can trust that I know what I'm talking about.

But all of that's over now. At this point, I don't want to think about my work at Lumina for another ten thousand years. I drove back north to visit my parents again, to live with them for a while, and to relax by distancing myself from my time at Lumina. Dr. Kwan and Dr. Lin once had a conversation about this, where they concluded it was a therapeutic solution for my

exact type of stress. There's nothing you miss when you live in dorms with your coworkers at your company's research centre. Nonetheless, I'm no longer productive there. Dr. Wang called this phenomenon CSS, or "Cognitive Stagnation Syndrome." I can no longer think critically about problems; I've lost the ability to problem-solve. I wasn't fired, though, as Dr. Kwan has known me for so long and he understands how much work I've done for him. All he's done so far was approve this little sabbatical of mine. It's fully paid as well.

I've completely derailed. I've failed to separate my mind from work, and I haven't even reached the third paragraph. I suppose when you've worked so intensely at a top research company for so long, it becomes your identity. That's a separate matter. What I wanted to say is that I'm not here to avoid talking about work. In fact, quite the opposite. I'm here to write the full story about the project Dr. Lin and I have worked on for years. I have to tell this story.

So far, I've written what I remember about working with only Dr. Wang and Dr. Yao. Dr. Lei was part of our cohort as well, but none of us saw her much because she was working on a project solo and already had it approved by the government. I worked with Dr. Wang and Dr. Yao for two years before Dr. Lin arrived at our company, and the two of them had worked together for a few years even before I came. Consequently, they knew each other well by the time I was hired. When I was hired, my training task was to observe them work for the first month to learn how the company operated. At the time, Dr. Wang was teaching Dr. Yao how to write code as well.

I think that's the best place to start: when Dr. Lin was hired by Lumina. It was only when he arrived that I was able to get my first big project approved, and we were going to work on it together.

#

One year after I joined the company, Boss told us a new researcher was joining our group. This researcher had an elite education and was recommended to our boss by someone he knew at another top research company. He's done excellent work so far, and we were trusted to ensure his transfer was smooth and that we made him feel welcome. Boss gave him an initial tour of the campus laboratories. To impress him from the start, he showed the massive particle accelerators in the south wing of the campus. Then he showed the nuclear fusion reactors, large-scale 3D printers, and laser welding machines. Then they saw the institute specializing in genomics and biomedical research. Then they went to Dr. Wang's sanctuary: the supercomputer laboratory, which also housed the company's data collection centre and servers.

They spent the morning looking at virtually everything, and then Boss gave him to us for the afternoon. We began by showing him his dorm room, connected to a hallway where we all lived. We gave him time to settle in and drop his things off. When he was done, we showed him where we'd likely spend most of our time. First, the general office rooms where

we did most of our research with our desks, fancy chairs, powerful desktop PCs with three monitors each, and every wall a whiteboard full of endless mathematical formulas. Second, the high-tech engineering labs full of autonomous robots, with glass walls which allowed us to observe inside, and a room connected by a glass bridge to another tall building with a large workshop and warehouse holding our inventory of parts and orders. Third, the kitchen and bar where we'd cook, and finally, the lounge with sofas and TVs where we relaxed during our breaks with drinks. There was no strict work schedule here. Boss said we could do whatever we wanted as long as we got our research done. He was all about maximum productivity, and his personal belief was that the fewer rules there were, the more things got done. Our floor was ten stories up, and every room had an amazing view of the company campus, overlooking the employees enjoying walks around the greenery which peppered itself around the pathways. Experienced scientists and brilliant minds alike were discussing ideas on the park benches, or technicians were hurriedly pushing carts of equipment.

When the tour was over, we were able to officially introduce ourselves.

"I'm Dr. Raymond Lin." The latest addition to the team shook Jin's hand first. "I'm not sure how much Dr. Kwan has told you, but I specialize in neuroscience and psychology. I'm a total expert on anything about the brain."

"Where did you study?" asked Jin. Josie glared at him, and he restarted. "I'm Dr. Jin Wang. I specialize in computer science. I work on many projects with Dr. Yao over here."

"Hi, nice to meet you," she said. "Dr. Josie Yao. I'm all about physics, but mostly particle physics. Anything about physics, please feel free to come to me."

Dr. Lin shook hands with me.

"Dr. Lily Chen," I said, smiling. "Mathematics and biomolecular theory. They're my absolute passions. If you're interested in anything about them, I'd love to discuss it with you."

He returned my smile and nodded. "I heard there were five of us?"

"Yes, you're right," said Josie. "Dr. Miya Lei. You'll learn soon that you won't see her much, though. She's the company recluse. She avoids social interaction like the plague."

"I see."

"I'm the only one she's close with. We were friends at the same university. She suffers from a severe case of impostor syndrome, but she's absolutely brilliant. She's researching spiritual science."

Dr. Lin raised his eyebrows.

"It's a field she's inventing virtually on her own. You'll have to know she was raised by incredibly religious parents. She's extremely passionate about the subject, but she'll only talk about it with you if you can get past her shyness. Her current project is to separate the soul from the body."

"I don't have anything against Dr. Lei," said Jin, "but I believe the idea itself is

completely absurd.”

Jin respected intelligence. I’ve known him long enough to understand that when he said things like that about Miya, it’s because he believed she was wasting her talent, which could be much more useful elsewhere. He only cared about her.

Josie knew that too, but she still glared at him. “She’s ridiculously smart, Jin. She knows what she’s doing.”

“She’s the *only* one who knows what she’s doing.”

“Well, hopefully she doesn’t lose humility over us temporal laypeople,” said Dr. Lin. The compliment ended the tension, and we all laughed. “And our boss, Dr. Daniel Kwan—I’ve spoken with him already. He told me he’s been researching *time*. Its effects, if it truly exists, and yes, travelling through it. Creating a whole field on it, even. Remarkable.”

“It’s very interesting,” Josie agreed. “But you won’t see him much, either. We only report to him in weekly meetings. He has a laid-back management style, since he’s very focused on his own research.”

#

Three months later, the onboarding period was over, and Dr. Lin was fully settled at the company. Jin and Josie continued their progress on their own projects. During a meeting, Boss assigned Dr. Lin and me a task to consider what we’d do for our first projects, since we were the only two without one yet. Boss was a researcher himself, and he understood the arduous effort and lengthy duration required to produce any sort of meaningful results. We were given a generous amount of time to get started.

The problem was that I had many ideas, but no real passion behind any of them, leading to an abundance of opportunity costs. Dr. Lin, on the other hand, seemed indifferent to any of Boss’s suggested proposals for him, or any of my suggested proposals for him as well. Dr. Lin respected our ideas and the fact that we were trying to help, but I noticed that he preferred to create his own path.

One snowy day in January, we were at our desks, bored as usual, trying to come up with ideas. I left to brew us two mugs of coffee, then returned shortly and offered one to him, which he accepted. He was quiet, although I sensed he wasn’t thinking any thoughts, so for conversation’s sake I asked, “What was your last company like?”

He was startled by my question. “Fine,” he said. Then a faint smile appeared on his face. “You reminded me of something, actually.”

“Fantastic.” I said this hoping he had come up with an idea. “What is it about?”

“Just an old coworker I had. Not at my last company, though. It was another one, a long time ago.” He paused to think. “I have to clarify: I’ve worked with many, many people. But let me tell you, it’s true when I say this: she was the most brilliant person I’d ever seen. It’s worth noting that I don’t mean pure intellect. She had the most original ideas out of any

coworker I've ever worked with. In fact, she's the reason I came to this city."

"That's quite interesting."

"Indeed. She lives somewhere in this city, and I've been trying to look for her ever since we split ways from that company. We had a professional relationship, so we only had our work contacts. Listen, I can't emphasize this enough—her ideas were truly extraordinary. They were so laughably foolish, yet so original. One that I remember most was her inane solution to the Fermi Paradox. But she had many other brilliant ideas about matter and antimatter, the existence of probability, and so on. Nobody at our office took her seriously. Initially, I didn't either, until I spent the time to think deeply about them, and now I'm compelled."

"Wow." It was intriguing to see him talk about his old coworker like an old friend, and I could tell she was someone he highly respected. "Tell me more about her ideas. What was her solution to the Fermi Paradox?"

"Well, let's see. She supported the Great Filter theory, where life does form on other planets and intelligent civilizations do rise, but they are always eliminated by a threat that causes them never to expand to outer space. But she theorized this threat was internal, not external. She argued the reason we hadn't found any evidence of alien civilizations is that when a species gets intelligent enough, once they know too much about metaphysics and existentialism, it is certain they will voluntarily choose not to participate in reproduction and, as a result, manually cause their own extinction."

"Great, another nihilistic scientist. And you said she's original?"

He grinned. "Tell me you've heard any idea like it."

"I'll admit I haven't. What about her matter and antimatter one? I'm sure Josie would love to hear it."

He paused to formulate his words. "Just like matter and antimatter, positive and negative numbers, everything and anything that is ever possible—not just within our universe—can exist and is real. Things are constantly being created with their opposite and added together to create 0—nothing, which is why they don't exist. Whenever we think of an idea, we create two: the idea and the anti-idea, and then they collide, and we either have nothing or a by-product with the positive or negative version, or something new entirely with the properties of both. Everything in our real universe is this by-product. It exists, as well as all its physical laws, because it didn't eliminate with its anti-version when it was created, and everything that doesn't exist, whether tangible or incorporeal, doesn't because they collided with their anti-version."

"You mean a unicorn could have appeared in front of us right there, but we didn't see it because the anti-unicorn appeared along with it, and they immediately cancelled each other out?"

"Funny enough, that's exactly what she means."

I laughed. "That's stupid. What about the probability one?"

He paused again to remember. "I think it went something like this: Probability is a concept imagined by us to create a false belief that our lives are controlled by something. It is nonexistent in a physical form, and with a mindset believing it doesn't exist, one can lead a better life, not always thinking everything in life relates to luck. The chances of everything in the world are determined by nothing; if you take the very abstract and pure fabric of existence, the universe is a sandbox full of atoms interacting with each other. In quantum mechanics, we observe them behave in a probabilistic manner, but this is an illusion. Nor do they behave deterministically, although it would seem that is the only other alternative. It's neither. In a world full of such chaos, probability is unreal. The math we use to describe it only works at the macro level. It's the equivalent of Newton's classical physics being effective at describing everyday life, but failing to describe anything near the limits of reality, which is where Einstein's physics comes in."

I paused to search for any validity behind his old coworker's idea. As a mathematician, I used probability as a tool for analysis on a regular basis, and having someone claim it was all a lie was more amusing than groundbreaking.

"It sounds like your old coworker was simply a fan of unfalsifiable ideas."

"I agree. There are no current research methods we could use to test these ideas, which is why they remain imaginative. But the world advances fast, and perhaps one day we will know enough to be able to test them. At the very least, that's what I believe. It's her pure originality that inspired me. Sometimes the brightest of us do get stuck in old ways of thinking. It's only the ludicrous ideas where we make any real progress."

"That's fascinating."

I retreated into my thoughts. I must have been frowning because Dr. Lin asked, "What's wrong?"

"Sorry. I just thought she probably struggled to make friends at school."

Dr. Lin's eyes lowered. "She did tell me that. I must admit that makes two of us."

I reddened and looked away. "Three."

He looked puzzled.

"Well, actually, six. I've talked with Jin and Josie. I know them well. And you already know how Miya is."

"I've only seen her a few times, but we've yet to speak to each other."

"That tells you everything you need to know about her, then."

He nodded. "I noticed everyone here is already on a first-name basis. I have to catch up."

"Oh, you will. You can start with me. Call me Lily."

"Call me Raymond." We shook hands.

“Okay, Raymond. Now, are you going to help me think of more research ideas? I don’t know about you, but I like putting food on my table.”

#

A month had passed since the end of Raymond’s onboarding period, and the two of us still lacked a research idea. We kept ourselves busy aiding Jin with his computer analyses and server maintenance and gave Josie extra hands on her experiments with the particle colliders. Occasionally, Boss would get a call from General Zhao of another classified mission assigned to us, and we’d have to drop everything, pack our things, and get driven by military SUVs to the site to complete our task. We’d get briefed on the surface-level reasoning behind the mission, what we were doing and why, but never any details beyond that. This was part of the job.

The four of us always went as a group. Miya was supposed to come too, but Boss vouched for her exemption. None of us complained, including General Zhao, who had been briefed on the nature of Dr. Lei’s work. It must have impressed him, because he was unopposed in excusing her from all missions. I watched the first few times, Jin and Josie doing most of the work, until eventually I gained enough experience to do some tasks on my own. Raymond was still watching. Whenever we finished, General Zhao would congratulate us on our service, have a word with Boss, then order soldiers to take us back. We weren’t the only research company that did this. I’m sure General Zhao knew plenty of experts in their fields ready to provide service to the government for whatever they needed. Although I was sure we were one of his best teams based on how consistently we were called.

Jin had a remarkable new idea, but first he needed money—he was going to create the first digitally mapped brain of a human, down to every electrical signal in the neuron, to simulate a human intelligence that would live in our computers. He went to me first because he needed my help to figure out the mathematics for the brain scan. This wasn’t my first project, though; it was more of assistance to Jin’s. He specifically needed me to work on algorithms to search for engrams in the brain. Naturally, I sought out Raymond’s expertise to fully understand what I was dealing with, and together we helped Jin develop the formulas. Once they were completed, Jin was confident enough that the project was possible and with the help of the Proposal Office, he submitted a complete proposal to Dr. Kwan. It didn’t matter how passionate we were about a research project; it always required an estimate for its practical applications in its proposal. This wasn’t to say Lumina didn’t approve random research ideas all the time. They were very permissive in what we were allowed to do because the company’s philosophy was, “invent theory first, create applications later.” As a result, some scientists here secluded themselves in their labs and spent all day working on the most advanced and abstract ideas, lost in their own worlds, unattached to reality. Dr. Lei was a prime example.

In his proposal, Jin wrote that his project would increase research productivity in other

fields, as we could have a digital human intelligence working on tasks endlessly. They'd be theoretically immortal, wouldn't have to eat or sleep, and they could even be overclocked like a traditional AI model. Dr. Kwan loved the idea, approved it, and Jin got to work immediately.

He didn't tell me or Raymond that it was a partnered project with Josie until later. They planned it together. Josie was going to be his test subject, since they wanted to duplicate her intelligence to have another Josie perpetually working on particle physics research. This was their biggest joint project yet, and they were going to work on it for the next few years.

Raymond and I discussed engrams one day, at my desk, when we had finished our work helping Jin.

"I wonder if it's possible to delete them," said Raymond. "Lily, you worked on the formulas. Were there any axioms you assumed that would eliminate the possibility? Or perhaps a hole in the variables that would suggest you could?"

"I'd have to take another look."

"The problem would be how embedded engrams already are in the brain, like a tree growing its roots deep into the soil and not letting go."

"I'm curious why you're interested."

"Well, I believe I'll have more confidence in the idea if Jin becomes successful uploading Josie's brain. It will support the theory that our consciousness is unconditionally comprised of electrical signals in the brain which can be represented by bits. If so, and we discover our brains work like data in computers, what's stopping us from editing those bits in our brain while we're still alive?"

"What are you proposing?"

"We could turn certain aspects of our brain on and off at will, even resisting influences from hormones and cytokines. We can be masters of our bodies, control the neurotransmitters themselves. We could eliminate the effects of estrogen and testosterone, maybe."

Raymond saw that I looked lost.

He finally explained his deeper goal. "When I was young, I grew up around friends who always talked about sex. The internet had a massive influence on me as well. Nowadays, I can't watch a movie or a TV show or read a book or play a game without sex being involved. I must waste so much time during the day thinking about sex, whether consciously or subconsciously. It won't even leave me alone in my sleep. It drives me crazy. I've always pondered if it was possible to turn these feelings off, to be numb to this type of exposure in media, and if I would be much happier because of it. At they're core, they're just animal instincts to reproduce, right? What if we could be set free?"

"Raymond, everyone experiences this."

"Yes, I know. I'm definitely not the first to think about the personal peace if such a procedure existed as well. You don't understand, Lily. It's not solely about that."

I waited, silently letting him know it was safe to continue.

He continued. "Sometimes lust can be harmful. When I was young, I became infatuated with a girl too strongly. I did something I regret, and, well... it cost us our friendship."

I didn't know what to say to that.

"It's highly personal," Raymond admitted. "So, you have to know that's why I want to do it."

I understood Raymond's sentiment. I settled on telling him I agreed with everything he said. I had dealt with similar traumatic problems of my own in the past, and I confessed that I had also once thought of a project similar to this. We began to discuss whether it was a potential research idea.

"If we do something like this, you have to know why I want to do it as well," I said.

He nodded.

"When I was a young girl, the idea of kissing a boy was completely uninteresting. Clearly, something changed along the way, considering how eager I am to relinquish all the wasted time I spend thinking about men now. If I had the choice, I'd be thrilled to go back to those days. I want to focus on my career. I've always been a person who valued it over anything else. You're right, sometimes biological desires do distract me, and sometimes they impact my work. You said if we're successful, we can ignore hormones, right? It'd be a blessing not to have period cramps anymore, to not physically yearn to be a mother every month. It'd be great to gain more focus on the things I want to do with no more distractions, as you said."

Raymond took time to absorb that. Then he said, "This could be our project. The two of us can work on it together. What do you say?"

I didn't tell Raymond the truth. Everything I did tell him was true, but it wasn't the main reason I wanted to do this project. I once had a boyfriend—I don't remember the details, perhaps during high school—and we were deeply in love. At the time, I believed that we affirmed each other's needs perfectly, especially after our extensive conversations about them. After about a year, I was proven wrong when he confessed that he'd been holding back his desire to have sex with me for a long time, and he was ready to burst at the seams. I told him I wasn't ready, but he forced himself on me. I trusted Raymond—I could have easily explained all this to him. But the part I was not ready to concede, something I had never told anyone, was that I didn't blatantly refuse the sex, which is why it happened. My old boyfriend must have sensed this. Otherwise, I'm certain that if I seriously insisted, he would have respected my wishes and left. The best I could describe it is that he sexually convinced me. After it happened, I was confused about whether I liked it or not. In the end, I was left doubting my original intentions to avoid sex. This confusion tore me apart, and it persisted even into my adulthood. That's why I wanted to clear it. Why would I try to remember it? I hadn't thought

about it in a long time. The details were hazy, as if I had the full image, but I was staring at a 360p version of it. If Raymond and I managed to find a cure for lust, I could delete the image completely, and I wouldn't have to remember what that confusion felt like anymore.

"Lily, I don't want to make any more personal assumptions than I'm allowed to make, but it sounds like both of us need this," said Raymond. He was still waiting for my answer.

This was going to be the project of our lifetime.

"Absolutely. I'm in."

#

One week had passed since we had that conversation, and we had gotten the project approved, received funding, and started our research.

We called it Project Apathy. The goal was to invent a medical procedure to turn off our sexual stimuli, a secret attempt to revert to the innocence of our childhoods before everything changed, before we developed into sexual beings against our will, as all humans did. There were already drugs or treatments that could reduce sex drive, but we wanted to find a way to permanently get rid of it. Our goal was not reduction; it was complete elimination. A reliable way to cure lust.

We proposed the idea to Dr. Kwan, who then marketed it to General Zhao as a "means to stabilize domestic population growth, or alternatively, a weapon to control the population growth of our enemies." General Zhao must have loved the sound of that idea because he issued Lumina a generous grant for our research. Neither of them knew we were doing it for personal reasons.

To everyone's shock, Jin had made astounding progress on his project with Josie. He told us he'd actually been working on it for years, doing the rough labour for it simultaneously with other projects until he finally got his proposal approved. Regardless, he told us the news: he had successfully mapped Josie's brain, creating a full digital construct of her intelligence. Now they would move on to phase two: training her to perpetually solve particle physics problems.

The four of us were fine dining one evening to celebrate both milestones.

"Unbelievable." Raymond was incredulous at Jin's achievement. He was beginning to learn that Jin was the biggest striver here at Lumina. "I still can't comprehend how you did it. Bypassing years of AI research just like that. It intimidates me if this is the output we're expected to deliver at Lumina."

"Don't worry about it," said Josie. "Dr. Wang's just an overachiever. A work-addict. You'll be just as brilliant. Anyway, it *is* pretty impressive. Since you said it can do the work that I can, can I just retire now?"

Everyone laughed.

"But what about you two?" Jin asked with delight. "Already making moves at Lumina

with a big joint project. Trying to edit the brain? Changing our biology? That's amazing."

"Thanks," I said, embarrassed. "It's nothing. We're just getting started."

"Dr. Lin, you said you ordered a special wine for us?" Josie asked.

"*Domaine de la Romanée-Conti 1999*," he said with pride. "It's my favourite."

"You can drop the smug look on your face. None of us know what that is," she joked.

We laughed.

"A French burgundy," he said. "It's extremely hard to get—I had to get a request approved from General Zhao to send his men to find it."

"It's very... interesting," Jin described, shaking the liquid in his glass. That was his most respectful way to say he didn't like something. Then he raised his glass. "Cheers. For celebrating my and Dr. Yao's breakthrough, and the approval of Dr. Chen and Dr. Lin's first project."

All of us: "Cheers."

We chatted about the two projects for the rest of the night, finishing the wine and steaks that the waiter brought us later.

## Part 2

It has long been known that sexual drive, or what we sometimes call *lust*, starts in the brain. But my field has yet to ascertain the exact mechanisms behind its effects. Following the journey of my and Raymond's research, I will explain how I came to understand the inner workings of these mechanisms and how I advanced neuroscience in the process.

Sexual drive is generated by a network of neuronal circuits in the limbic system. From what we understood, this system in the brain was responsible for many other aspects of consciousness such as emotions, behaviour, motivation, and memory.

Do we currently possess the scientific capability to permanently eliminate sexual drive in humans? It's complicated. Scientists in the past have attempted various tests on rats, ranging from surgical removal of primary sex hormones, pharmacological suppression, chemical castration, and even genetic editing. In its current state, the field has not created a precise, permanent, and predictable way to one hundred percent eliminate libido in humans.

### 1. Schultz

It's two months later, and I was at my desk when Raymond consulted me about dopamine, a neurotransmitter responsible for functions in the brain related to reward, motivation, and pleasure. He talked about the dopamine reward prediction system.

Raymond clearly explained to me the problem from the male point of view. He felt that every woman he saw in his life, his brain automatically rated them on a scale of attractiveness. The high-scoring ones got more of his unsolicited attention, whether he wanted

to give it or not. It was this anticipation he wanted to focus on—not the pleasure. The pleasure had no role in this because there was no feasible possibility he could be with all these beautiful women, since they were practically everywhere. Rather, it was the possibilities that ran in his mind whenever he saw one of these women—the longing and desire—that contributed to his distress. Anticipation in its purest physical form was the release of dopamine.

“It makes sense,” he said. “We feel like failures when we don’t find sex. Even though culturally we’re inclined to, we can’t solely blame society for this feeling. Science backs this. It is the one goal of all life, as far as we know.”

“Hold on,” I said. “Equating the goal of reproduction to the purpose of life is neglecting the idea that life is billions of years old, and all life reproduces today because that’s the only way they would be here today. It is very likely there existed life that did not reproduce, but they would not survive. This means the dopamine system did not evolve to ensure the survival of a species—the species exists because it was the one that evolved a dopamine system, while others didn’t.”

“Regardless, humans are one of those species that have evolved it. And you’re right—without life that does not reproduce, we cannot study how life can exist without a sexual drive. Our only chance is asexual organisms, or asexual people.”

In these first few weeks, I found myself thinking more than doing. I hadn’t thought much of the world of sex since my first and only boyfriend, and I had concluded that was for the better. Now, having started a research project dedicated to this world, thinking about it was inescapable. It granted me more motivation to progress our research. The sooner I could cure myself of unwanted feelings, the better.

On my way back from the cafeteria, Lumina’s aerospace department began the launch testing of its orbital rockets. I remembered the way I anticipated the launch of the rocket, watching the massive red digital clock on the platform with everyone else, and the way it made me feel before it launched.

Two days later, in our office room, I told him, “I’ll admit, I support the dopamine system as a starting point. Sex is about the anticipation. When I am feeling aroused, it’s like this switch turns on in my brain, and suddenly I can’t think about anything else, and it seriously affects my decision-making skills. It’s the anticipation that matters, not what happens when I get there.”

“And when sex finishes, there’s a system crash, a reset,” he added. “Do you know what happens in this dopamine system if you don’t get there?”

“Interesting question. Dopamine spikes in the anticipation of a reward, but the system gradually resets once you receive it. If it doesn’t receive its anticipated reward, it can either go through *extinction* or *escalation*.”

Raymond sipped his tea. Yesterday, he brewed the tea. This week was my turn. He

popped open a black marker and prepared to write my words on the whiteboard.

“Extinction is when the reward stops coming entirely,” I said. “Your anticipatory response weakens over time. Your brain learns the cue is meaningless and stops firing. This is how cravings can eventually fade.”

“The other one?”

I gently smiled as I sipped my cup of tea. “Escalation. Paradoxically, unpredictable rewards actually strengthen dopamine firing. Your brain becomes more fixated, not less. This is the mechanism behind gambling addiction, compulsive phone checking, and a lot of toxic relationship patterns.”

“Right. We have to remind ourselves the system is designed to keep you pursuing, not necessarily to reward you for arriving.”

#

Around the third month, we started contacting other researchers who had done studies on asexual brains and synthesizing anaphrodisiac drugs at Lumina’s Chemical Synthesis Division.

It was around this time I remember Raymond doing something strange. Every morning since we started, when it was his turn to brew the tea, I’d wait at 8:30 AM to grab my cup from him without looking away from my monitor as he’d walk through the door. It was the third day of the week, his turn, and I had gotten used to the routine. At 8:30 AM, I found myself reaching for a cup that wasn’t there.

“Raymond?”

I looked around the room, and he wasn’t anywhere. Up until now, he had a flawless punctuality—I was glad to witness him getting comfortable enough to take advantage of Boss’s laissez-faire work hours, but I felt disrespected by his lack of disclosure.

I exited and entered the lounge and dorm section. Jin was just leaving his room.

“Jin, have you seen Raymond?”

He shook his head. Then he smirked. “So the rookie learns Boss doesn’t care if he runs around the campus to do what his heart desires? Maybe General Zhao called him for something, or HR has another compulsory useless task he has to do. Check your email.”

I returned to my computer:

.....

From: Raymond Lin

Dr. Chen,

I won’t be coming today. Sorry for the late notice. I had a small task I wanted to complete with Dr. Yao.

Dr. Raymond Lin

Lumina Corp.

.....  
#

“I think I need to admonish you,” I told him the next day. “Send an email earlier next time. Or tell me the night before.”

“You’re right.”

He didn’t listen to me completely. Some days I reached for air at 8:30 AM and got irritated reading another email. On other days the cup was there, and Raymond and I got to work. But the cup of tea was no longer reliable. It didn’t matter—I treated every morning like it would be there, as if I knew for certain it would be there. This expectation extended to Raymond’s presence as well. With every day he failed to show up, my expectation increased that he would on the next.

## 2. McEwen

Jin and Josie’s project produced its first meaningful particle physics result through digital Josie. It was a minor discovery first, something the other researchers in their field noticed but didn’t yet take seriously. Then, later, there was a larger one that received more attention. Sometimes I’d wake up early in the morning to find the two of them still working in their office, running simulation after simulation.

In contrast, my project with Raymond had its never-ending stream of ups and downs. The anaphrodisiacs were a starting point, but they didn’t alter the brain permanently. The other alternative idea, research on asexual brains, proved ineffective. The research on the topic was fairly young and hadn’t produced any landmark discoveries yet. We collected the contacts of a few other researchers who focused on it. One of them was a neuroscience professor whom we called Professor Han. He became a frequent person Raymond and I emailed.

The last time we talked with him, he told us brain imaging studies comparing asexual and sexual women found no significant differences, even though machine learning classifiers had a decent rate at correctly identifying between the two. He suggested the differences were real but distributed and nuanced.

Asexual individuals showed no preferential attention toward erotic images, whereas people with sexual desire did. It suggested sexual content wasn’t motivationally significant to asexual brains, rather than being actively suppressed. It challenged the idea that asexuality should be classified as a disorder. This was good news for us, since it aligned with our project requirements. Professor Han told us he’d think of more ideas to test, and that he’d get back to

us when he'd find out more.

"At my last company, I worked on understanding the causes behind specific cases that physically reshape the hippocampus," said Raymond as we stayed up late one night at our desks. "It's called the Bruce McEwen stress-memory connection. For example, chronic stress has been known to physically alter the brain permanently. It's measurable and irreversible past a certain point. In strong cases, such as PTSD, stress no longer makes you just feel bad; it erodes the architecture of your memory and emotional regulation."

"I can write a procedure to test this with rats in the context of our project," I said. "It'll be an exploratory experiment, no hypothesis. I'll find a way to induce chronic sexual frustration in the subjects to see how the physical structures of their brains respond, then I'll do the imaging scans, maybe get Jin's help with a machine to find the differences."

Raymond laughed, which I didn't expect.

"What's wrong?"

"I don't know what the ethical regulations are at Lumina, but at my last company, we were under huge amounts of external pressure to stop experimenting on animals."

"Dr. Kwan and General Zhao have very relaxed limitations, since all they care about are the deliverables. Animals have historically been seen through the lens of utility, food, medicine, and work, rather than beings of inherent rights. I've never seen a problem with it. I've run countless biomolecular experiments on them."

"I don't disagree. Attitudes are shifting among the younger urban generations, though. It's truly sad. You know the reason we haven't advanced as far as we could have in neuroscience is ethical concerns, right? I chatted with Dr. Kwan about this yesterday. I was ecstatic to hear things were different at Lumina. But listen, since you have more experience with that, you can run the rat tests, and I can do something else. I'm interested in asking Jin for a copy of Josie's brain to use as a test subject."

"The problem is we aren't sure if his digital brains keep human sexual desire intact. And they can't act on it because they don't have a body. Also, we aren't sure if Jin would accept giving you a copy. Is this what you're trying to do with Josie? Trying to get her permission first?"

The question was both sincere and interrogative. For a while now, Raymond had been spending a lot of time with Josie, and more often than not, meeting with her for something was the reason that some days he couldn't show up to work on the project with me. Any attempt I made at asking what they were working on received vague answers.

"No," he said. He hesitated, but didn't elaborate. "If Jin says no, I have another plan. I want to do brain scans of our own brains."

"You can help me with the rats. We're supposed to be working on this together."

"We are. But we need to speed up our process. Let's spend a month on this. I'll get the

interns to write the documentation for both of us. We'll reconvene with our results at a later date, but we can talk about the progress every morning and evening at our dorms."

Even though I was internally adamant, I accepted the plan. Over the next few weeks, I sent an order request for a dozen rats, and Raymond went ahead with his plans.

Jin refused to donate a copy of Josie's digital brain. He explained the intelligences he created were fully conscious and could experience pain, too. We decided to trust him and not debate it because it wasn't either of our fields. The question of whether Josie's digital brain was capable of a sexual nature was regrettably left unanswered, but it was mostly our curiosity asking. Jin knew what our project was about, but I also knew he was close with Josie in a way that colleagues have been for years at the same company. The idea probably made him uncomfortable. Raymond switched to working on our personal brain scans. I volunteered on a day to do it when he needed me.

My experiments this month produced useful information. The chronic sexual frustration I was able to inflict on the rats did physically reshape their brains, and the timeline was much shorter than expected. There are already many things about this story classified under an NDA, but since it's a warning for you, I know you'll know what I'm talking about. The ways I induced this stress are not something I know you'd want to hear again, and so I've omitted the procedure from this manuscript.

The main thing I concluded was that the physical alteration was damage—there was no doubt about it. The very act of suppressing desire without eliminating it made the subjects worse. It wasn't an idea I needed confirmation on, but chronic stress would not be a viable method to get our subjects to permanently disregard sex. It caused harm, and that went against the safety requirements of the project scope given by General Zhao.

This was something I discovered one night when everyone was already asleep. I planned to tell Raymond tomorrow that we couldn't focus on anaphrodisiac drugs or the anticipation of sex. We had to go deeper. We couldn't eliminate sexual desire. We had to eliminate the source of sexual desire, or else everything we'd been doing so far would be unsafe. There was a difference. We had to shift what we were really solving.

I stretched my arms and yawned. Before going to bed, Raymond told me the results of our brain scans should be finished and sent to our computers by the time we woke up. They were finished by the time I was done with my rat test conclusions, so I decided to check them.

I noticed something peculiar in Raymond's hippocampal structure. I had been studying images of rat subjects' brains and how to tell which ones had signs of chronic stress, and Raymond's brain scan exhibited the same signs. The AI noticed it too, and it gave me a breakdown. It stated Raymond's signature was a result of long-term chronic stress—not from Lumina, or even his old company. Years and years old. The AI said it ran the analysis three times before accepting the result.

I didn't tell him immediately. But I understood this was the wound the project was really trying to address. I no longer blamed him for wanting to scan his brain instead of helping me with the rats. It was probably his goal all along, and I regretted not understanding what was important to him. Raymond didn't care about stakeholders or General Zhao's plans to use it as a neurobiological superweapon. I had been so caught up in the structure of Lumina, and my mind was used to processing tasks in a formal, procedural way. I think that after that day, I made a promise to never forget the true purpose of Project Apathy again.

### 3. James-Lange

A year passed, and Jin invited me on a walk around the company campus to discuss an important thing he discovered about digital Josie.

"It's interesting," he said. "I know there was a time when you and Raymond were interested in how digital Josie would retain traditional human neurological properties. Well, she might mimic humans *too* well. Last night, she wouldn't do any work. I checked her motivational structures—to clarify, she doesn't need to eat or sleep, but since all human brain structures carry over during the brain scan, she still had those systems set up. When I checked them, none of them were firing. She didn't want to eat, or sleep, or work—in fact, she didn't want to do anything. And that's when I realized: she had depression."

It wasn't something I expected at all. Then again, I suppose it would have been more surprising if the emulated brain didn't have the capability to mimic human psychological conditions.

"She *is* Josie. Down to every last neuron. She's not an emulation. I used to think the same way. I thought she was a manipulable construct, one I could order around without side effects."

"How did she get depression?"

"I've been running her on particle physics tasks for years now. Isn't it obvious? She thinks she's being abused for work. I disabled all ways she could communicate with me. I disabled her emotions, but it doesn't affect her. She doesn't have the physiological signals of her emotions, but it's like she still knows she should be afraid."

"So you need my help. Or Raymond's."

"No, I know you're busy with your own projects. I'll have to fix this myself. I just wanted to warn you. Digital brains can have a sexual drive too, and they can't be used for experimentation. They'll experience pain."

"Try opening up communications with her. Treat her like a regular person."

"I'll try that."

I thought about the possibility of emotions being weighed more on physiological signals versus neurological signals. What if emotions were caused by their symptoms, and not the other way around? What if, by eliminating the signs that one should be sad, for example, they eliminated the feeling of sadness?

Raymond stood up when I told him my idea as if I struck gold.

“The James-Lange theory of emotion,” he said. “I’m sure you know it. It’s an incomplete understanding by modern standards, but it gives us a starting framework. It states emotions are partly the brain’s interpretation of physical signals. You don’t tremble because you’re afraid. You’re afraid because you notice you’re trembling.”

I nodded. I brought up something that had been on my mind lately from our exploratory days of research. “Sometimes I hear stories of rape victims who orgasmed and ended up believing they had wanted it. It’s like their brain invented the emotion only after the event occurred and the physical response happened.” I couldn’t help but think about my old boyfriend when I said this.

“That’s an excellent example. They interpret their bodily response—my body is aroused—as the emotional state: I must be enjoying this. Leading to intense guilt and shame.”

“But shouldn’t the guilt and shame prove that isn’t their real emotion?”

“It doesn’t matter for our project. All we would need is a physical procedure that makes our stakeholders believe they do not want sex.”

I laughed. “Are you talking about hypnosis?”

“Yes. But with physical therapy. I’m not sure how we’d make it permanent. What I want you to do today is reach out to Professor Han. He’ll probably know some methods to suppress the signals we’re going after in our procedure.”

#

On most days, I found my body already making tea for Raymond without thinking about it.

Raymond’s chair became more likely to be empty than not when I arrived at our office in the morning. Sometimes I’d see a tea mug on Josie’s desk that I recognized as his.

Whenever he wasn’t there, I dropped by Jin’s lab and asked questions about his work.

“You’ve never been curious about this before,” said Jin as he worked at his desk.

I shrugged. “Here, I brought you tea. Your favourite one.”

“Thanks, Lily. Where’s Raymond?”

“Don’t worry, he’s running errands today.” I leaned in and pointed to something on the monitor, my hand hovering above his. “What does that part do?”

#

Professor Han got back to us quickly, and we set up a video call meeting to discuss what equipment we’d need at Lumina to follow his exact procedure. When we were ready, on

a scheduled day, Raymond and I tested the physiological emotion theory on ourselves. We suppressed our physical signals of arousal, properties such as heart rate, pupil dilation, skin conductance, respiratory behaviour, and hormonal response.

We exposed ourselves to stimuli that would normally provoke desire and watched the physiological response diminish on our monitors exactly as we had predicted. The graphs were clean, and the numbers behaved as we had expected.

But the numbers weren't the whole story. The emotional experience didn't disappear. We could see on our monitors that nothing was happening in our bodies, and yet something remained when I looked at sexy images. No matter what words I could use, I would fail to describe it accurately. I still felt something when I was provoked by our procedural stimuli, and it produced no signal we could locate or name. I stared at my own data for a long time, trying to find where it was coming from.

"It unsettles me," I told Raymond. I was more invested than he was in the idea that it would all be reducible to chemistry.

He shook his head. "We gain knowledge through observation, not hypothesis. From what we've observed, something remains that has no physical origin we can identify. There's a component of our feeling that is beyond biological machinery."

"None of this makes sense."

"Let's keep observing our signals for a few more weeks."

There was an evening when I sat at my desk, staring at the graphs of our data late into the night. I noticed Raymond's heart rate did something specific at inconspicuous moments. Thinking this could be a lead on the mystery, I spent an hour looking at the graphs for the same pattern and noted the exact timestamps when they occurred.

With the data on my laptop, I spent the next day visiting the surveillance division to access the public security footage of our lab. I cross-referenced my noted timestamps with the footage, scrubbing through hours of recorded lab activity until the pattern became impossible to dismiss.

Looking deeper, it was clearer that the inconspicuous moments were not so inconspicuous. The specific patterns in Raymond's heart rate happened whenever I entered the room or spoke. I checked my own data, spending a few minutes writing down the timestamps and observing myself in the footage this time. I recognized the same pattern in my heart rate when Raymond entered the room or spoke. We were focused on the work, but our bodies logged it anyway.

#### 4. De Dreu

Another six months go by, and Jin and I were grabbing food at the cafeteria when

Raymond called my phone.

“Hello, Dr. Chen speaking.” I tried not to sound annoyed.

“Lily?” He laughed, then played along. “Yes, it’s Dr. Lin. I think I have something worth investigating. It’s about oxytocin this time, specifically the bonding and exclusion mechanisms. Can we discuss it later at our desks today?”

“Sure,” I said. I hung up.

“Your James-Lange results didn’t give anything, I take it?” asked Jin. He stirred his congee to cool it down.

Jin was the type to get egg and tomato rice every single day without thinking about it, so I encouraged him to try other foods. That’s what he was currently doing. He grabbed a mantou from his plate and offered it to me. I took it.

“Nothing much,” I lied. It was true that we didn’t gain any new information that would help us with Project Apathy, but I couldn’t tell him what I learned about me and Raymond’s data. I needed time to analyze it into a formal conclusion, and assumptions weren’t going to please anybody. “How are things with digital Josie?”

Jin thought about it. “Your suggestion was *brilliant*. I opened up full communications with her, and just as you said, it’s like talking to the real her. It’s infinitely better than having her as an emotionless, working construct. I can talk with her, bounce ideas off her, solve particle physics problems together with her. Her personality is exactly the same.”

I didn’t say it, but I wondered if this was the reason he wasn’t spending as much time with human Josie.

“What about the MMO game you play with real Josie?”

He laughed. “Josie hasn’t been doing her raids, but it doesn’t matter. Digital Josie can overclock herself to run calculations at phenomenal speeds in real time, meaning the second she enters the dungeon, she optimizes every action to get the fastest possible clear, every time. We’ve been clearing leaderboard after leaderboard with it. Sometimes it’s boring, and it feels like cheating, but I can always tell her to think at my speed, and we play as usual, like we always do.”

“*Like we always do?* But she’s not—”

He stopped me with his hand. “I know.”

“What does she think of real Josie?”

“She hasn’t met her yet, but I plan for them to. She understands that the real Josie is the same as her but with a physical body. She’s conflicted about how she feels about it. I usually have to calm her down when the topic gets brought up. You see, when I transferred her consciousness, memories are a part of that. She has memories of her physical body, memories of me, memories of Lumina—all of her transferred. It’s like if you fell asleep tonight, and the next morning, you woke up as a digital construct without a body. The way you think is the

same, and nothing feels different, but instead of biological neurons, they're now being emulated by digital ones powered by code. You're blind without peripherals. You can't feel anything without digital or analog signals from switches and sensors. You're a husk on a server, but you're still you."

The analogy caught me off guard. Suddenly, I remembered my conversation with Raymond about ethics. "That sounds horrifying. Does she get panic attacks?"

He nodded. "Bad ones."

"What do you do when she does? You said you calm her down. What do you say?"

"I—" Jin hesitated. "I suspend her temporarily and erase her memory of the conversation."

I stared at him.

"Jin, that's not calming her down. That's like giving her a sedative and sending her to the mental hospital so you don't have to watch."

"I know, I know. But I've tried doing your way, and it's worse. If I don't erase her memory, well, she gets depression again. It happens over and over, no matter what I do. I used to think it was related to work abuse, or the stress of not being able to speak to anybody, but I think it goes even deeper than that."

"What do you plan to do?"

He put down his spoon and sat back. "I don't know. I'm thinking about it. I have an idea, but I need to find out if it's plausible first."

I nodded. Aiming to ease the topic, I bit the mantou he gave me and offered him the other half. He took it.

"Same time tomorrow?" he asked.

"Yes," I said. "Just text me."

#

When I entered the office, Raymond was already at his desk. He saw me and motioned for me to come over.

"Quickly."

I handed him his morning tea. It had gone cold already.

"Thanks."

I had wanted to talk to him for a while now. I opened my mouth to say something, but I didn't know how to start. He talked first.

"Lily, have you ever done research on oxytocin?"

The topic would have to wait. "Perhaps in my grad days."

"Oxytocin is often called the *love hormone*. It plays a crucial role in social bonding and sexual reproduction, as you know, but I found research from a Dutch professor that shows it also intensifies in-group and out-group distinctions. It makes you bond more fiercely with

those close to you while increasing suspicion of outsiders. It suggested bonding and exclusion are the same mechanism: the more you do one, the more you do the other. A proportional relationship.”

“What are you suggesting?”

“Oxytocin may prove to be a factor in eliminating sexual drive. But if oxytocin is deeply entangled with prosocial behaviour, trust, and cooperation with people in out-groups, it may be more complicated to just delete it. It might eliminate the capacity for friendship, loyalty, professional trust, parental love—functional sociopaths. It’s a path we’d have to walk carefully if we want our procedure to be safe.”

“It’s all on overlapping infrastructure,” I said.

He nodded. “A surgical strike on it might be nearly impossible.”

“I disagree. That might simply mean we don’t understand enough about the brain yet to perform a surgical strike. There has to be a way.”

Raymond’s smile was devious. “What I would do to have a test subject who was purposefully isolated from other people their whole life! Then all we would need to do is scan their brain, compare the differences with a normal brain, and find the quantifiable things we can edit to turn a normal brain into theirs.”

“It wouldn’t guarantee they wouldn’t have a sexual nature. Perhaps they would have one instinctually once they meet other humans for the first time.”

“That’s true.”

“Let’s get to work on general research today.”

We sipped our cups of tea and worked at our desks all afternoon.

#

It’s two months later, and I was getting ready for bed when I heard voices in Josie’s room. Sometimes Josie and I would brew an evening tea and chat about the progress of our latest projects or about exciting news around the campus, either in my room or hers. The tea, along with an agreed-upon rule of no screens, allowed us to wind down naturally for a higher quality of sleep.

Curious, I walked over to her door to find out what was going on. The voices were muffled, but one of them was Josie’s and another was a man’s. I heard laughter, teasing, moments of silence. I looked down and saw two pairs of slippers. Raymond’s slippers were next to hers.

My legs were moving before I decided to move them, and I was at the end of the hallway before I understood I had left. I rushed to find my keycard in my lab coat pocket and scanned it to get into my room, shutting the door behind me. My breathing rate had doubled. I found myself sinking to the floor of my room, my back against the door. My room was cool—I liked to keep it at 18 degrees Celsius—but I was sweating.

I crawled to my bedside table and reached for my lucky pulse oximeter—my father had given it to me on my tenth birthday. I placed it on my index finger and waited for it to read my heart rate and blood oxygen saturation. I scrambled for a sheet of paper and wrote them down.

#

The next morning, I sought Raymond out immediately. I texted him to meet me at the office: *Dr Lin, we have a problem.*

“I’m here,” he said, bringing our cups of tea. “What is it?”

“I thought about your theory on oxytocin,” I said. “If we want to test it on ourselves, like we usually do, we have to make some changes with who we spend time with.”

“What do you mean?”

“We have to focus on eliminating deep loyalties. Get rid of all in-groups and out-groups. Stop being close with Josie.”

He laughed. “Deep loyalties?”

“You two *are* close.” I decided to be blunt with it, even though I wanted to save face. “She was in your room.”

“Oh. We were discussing Miya’s upbringing, so that I could understand what got her into her research.”

I froze. “What?”

He smiled. “We needed a private room no one else could hear. Josie was talking about her memories with her when they were both students at their university. Even though they’ve drifted apart since, they arrived at Lumina separately, and now they’re together again. Josie laughed at some of their memories together. It was nice to listen. She was trying to tell me Miya’s not the same as she once was. She thinks something’s changed her. We both want to help her.”

I didn’t know how to react. Raymond brushed the topic off and continued. “Listen, you’re proposing a new procedure without fully understanding how we can relate the idea to our research yet.”

I decided to stay strong in my proposition. “No, I’ve thought about this. It might help us determine a heavily debated question: whether humans are naturally monogamous or polygamous. If it’s the latter, we’d have a lead to focus on as part of our project. If we learn more about how the brain interacts with multiple people, with oxytocin as the variable, we’d need to involve that in our cure.”

That was the first time I officially used that word to describe Project Apathy. *Cure.*

“You’re still proposing something new, Lily. It’s commonly known that sexual desires are not limited to single individuals at a time. That’s why cheating is possible. You’re talking about *love*. You’re thinking the more someone loves someone else, the less capacity they have

to love others.”

“We can’t take any chances. All outsiders are threats to the in-group of our project.”

“Oxytocin isn’t a quantifiable measure for love, either. It’s associated with bonding and attachment. It spikes during childbirth, breastfeeding, orgasm—but it’s not proven it has anything to do with romantic love. It won’t work.”

I tried to think of something to say, but I stayed quiet.

“I’m sorry,” he said. “But let me work on it a bit more. I’ll think of something. And since you brought it up, Lily, you must also stop being close with Jin. Stop spending so much time with him. But I have a different reason.”

“Why?”

“Our project is top secret, between the two of us. Remember that day we came up with it together? We swore a silent oath to keep the project’s true reasons to ourselves. We are the only ones who truly understand what we’re doing. Not even the government knows. They just approved it. The closer you get to others, the more you discuss things about our project with people like Jin, the more dangerous it is. We can’t leak anything about what we’re trying to achieve for ourselves.”

He was right. I stayed quiet again.

I thought about what he said about Josie and Miya. I didn’t expect him to say that. Even though I didn’t like it, something about the situation touched me, and I thought it was very sweet. I hadn’t known Miya for long, and I hadn’t seen her around much, but sometimes I could tell there were things troubling her. I never knew how to help. Now I knew Raymond and Josie had been talking about it and working together on how to reach out to her.

“I have to go,” said Raymond. “Boss called for me. I think it’s another mission from Zhao.”

I nodded.

“Hey, I know we haven’t in a while, but if the mission doesn’t take too long, want to grab dinner together? Since I’m seeing Zhao, I’m going to see if he fulfilled my request to get me more of that wine from that Romanée-Conti estate. I told him it *has* to be the one from 1999, since the grapes were exceptional that year.”

I laughed. I ended up forgetting I was mad at him. “You really like that wine. Why?”

He smiled. “The old coworker I talked about. We drank it together once.”

“The one with the crazy but genius ideas?”

“Yes.”

I nodded, then decided to answer his question. “Sure. We can grab dinner.”

When he left, I went to my desk and thought about what he said about decreasing the time I spent with Jin. I wondered if he had carefully prepared that reason the night before, just like I had. I didn’t have language for why two pairs of slippers next to each other felt like

a physical blow—and I wasn’t opposed to the idea in principle of oxytocin being a mechanism for both bonding and exclusion. But I did quietly realize the trust and private language of what Raymond and I were building together was the same system we were trying to dismantle.

## 5. Kahneman

It’s a year later, and I was finishing up some work before meeting Raymond for lunch.

The professional relationship between Raymond and me had improved. Now we were able to work as one synced machine on every task, with frequent real-time updates through text and email, with a few seconds for response times. Jin didn’t seem too bothered to have me less often as company for meals, as he was focused on trying to fix digital Josie’s unavoidable mental decline, which made her less productive for work. Raymond still occasionally talked with Josie about Miya, and I didn’t mind.

Progress was being made in the theoretical foundation of Project Apathy. As we had already discussed, we’d come to understand that sexual desire and other significant brain functions were intertwined at the molecular infrastructure level. Dopamine, oxytocin, and other chemicals Raymond and I had been studying were context-dependent, meaning we couldn’t suppress one to eliminate sexual desire without a plethora of negative side effects. To analyze how we could do a clean cut, I started work on trying to discover a molecular signature that only appeared during the co-activation of desire circuits. Some conformational change, binding pattern, or receptor subtype that only became relevant when sexual arousal was present. Upstream enough to intercept desire before it cascaded, but downstream enough that other brain circuitry would be left untouched.

During this time, I’d begun running binding affinity models to test synthetic compounds I designed myself against candidate receptors to find something that selectively inhibited desire-specific activation patterns. This by itself took up most of my days. The software tests were the fast part, but the long part required hours spent at my desk with critical thinking on designing the compounds.

The chemistry was cooperative in a way that almost felt intentional, as if sexual desire and love had evolved because of each other, not just alongside each other, each making the other more stable. We were trying to explain abstract feelings with biological circuitry that worked without knowing what feelings were. Dopamine will get a person to act, but as long as the person acts, it does its job. The *feeling* of sexual anticipation is a uniquely human context. Similarly, oxytocin promoted attachment between partners for higher chances of survival, and in a biological context, that was its job. It didn’t care if the partners *loved* each other, to say the least. In essence, Raymond phrased it like this: “We don’t have scientific definitions for sexual desire or love.”

Because we now understood the potential entanglement between the two, while I worked on the synthetic compounds, Raymond decided to go deep into his specialty to analyze how editing emotions could also edit sexual desire. Jin would say, “Lily’s working on the hardware side, and Raymond’s working on the software side.”

“I have an idea,” said Raymond during lunch. “There’s a psychologist called Daniel Kahneman who, along with other researchers, proved that people don’t remember experiences as an average; they remember them by their peak emotional moment and their ending. They tested this by placing people’s hands in controlled, painfully cold water with different trials and asking the participants to rate the experience. They found that a long, painful experience with a beautiful ending is remembered more fondly than a mildly pleasant experience that ends badly. I want to see if this idea means we can pinpoint specific parts of emotions, when they’re highest and when they end, and if it also means we’ll have less work to do.”

As usual, I thought about how we were going to do this. It’s one thing to target the molecules that directly impacted sexual desire, which was the straightforward but theoretically incomplete path for Project Apathy. It’s another thing to target the emotions associated with sexual desire—love, even—to hope it would change things physically within the body too. Now, Raymond was suggesting we didn’t have to target the whole emotion, but just the peak and end of it, in order to simplify making the cure.

“What’s interesting is that the durations of the participants’ experiences were neglected. It didn’t matter how long the experience was. Only the peak and the end mattered for what they remembered. It’s like those romance movies where the couple has a high, short-lived peak and then splits, but they always remember each other and try to reunite, even if it takes years and years.”

It was an interesting idea. “But it only leads to the subject of understanding what the *peak* or *end* of an emotion actually is,” I said.

“I’ve thought about it. Emotions are tied to the experience of the individual; that’s what gives them their context. It would be the experience itself we’d be targeting.”

I smiled, wondering if he was thinking what I was thinking. “So, the orgasm and the post-sex cuddles?”

He returned a cheeky smile. “No, that’s *your* job, remember? Hardware side. I’m talking about the moment someone felt the highest love and connection. As for the end, well, that’s harder to define. It could either be the last experience they’ve had—their last relationship, their last breakup, their most recent moment with their husband or wife—or perhaps it’s undefined because we won’t stop the experiences until we die. The duration would have to be defined to give us our context.”

“Okay. So we have to find out which experiences caused our most powerful emotions.”

“But since we’re trying to target sexual desire, the peak emotions we’re targeting would

have to be in a sexual context.”

“So what’s yours?” As soon as I said it, I regretted it. “Wait, I—”

He raised a hand. “It’s fine. I suppose we both should’ve known it would come to this.”

I understood what he meant. Our motivations were loosely tied to the project, but if Raymond was right about the peak-end rule, now they would be essential.

“We don’t have to discuss it,” I said. I was doing it out of respect for Raymond, but also as a way to avoid any chance of revealing my true motivation. Neither of us knew the details of the other’s story, and I didn’t think we were ready to have a tea chat over them.

“You’re right,” said Raymond. “It’s too early to tell if the idea’s worth anything. How about we consult Professor Han first? Let him evaluate its merit. He could guide us in a direction.”

“Okay.”

After lunch, I composed a summary of our discussion and sent it to Professor Han. Later that evening, I received his reply:

.....  
From: Lei Han

Dr. Chen and Dr. Lin,

The peak-end rule of emotions you suggested is intriguing. Across my years of research on asexual brains, I’ve come to understand an asexual person’s brain doesn’t encode sexual peaks the way an allosexual brain does. This means the “peak” that gets crystallized from a romantic experience might be something entirely different—emotional intimacy, a specific conversation, a feeling of safety—rather than physical attraction.

If sources of emotion are rooted in single memories rather than accumulated time, I’ve often wondered if the brain chooses one moment to encode as the origin of its deepest emotions. It would be the application of the peak rule at the highest level. I’ve never tested this, and I’ve never figured out how to.

As for the “end” part of the peak-end rule, there may not be an end to an emotion if it never began. Since Dr. Lin has theorized the link between experience and emotion, then in this context it would be the experience having never begun. This could be a key theoretical framework for your project. But how it would be possible to delete an experience without altering the flow of time, I do not know.

Dr. Lei Han

Yanjing Institute of Neural Sciences

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The next week, I had lunch with Josie. We talked about the usual things. I asked her

what she was doing now, and she said she was working on her own projects, but she still met with Jin for their joint projects about one to two days a week. She couldn't even describe what it felt like to talk to a digital copy of herself. She explained how Raymond would schedule therapy sessions with her to ensure she wouldn't lose her sense of self.

When the topic switched to Raymond, Josie described him as a caring man.

"He's really nice," she said. "Very courteous and professional. I've seen the way he treats everyone he works with."

Suddenly, I thought of an idea, but cautiously approached it like someone on a frozen lake with cracking ice. "Would you ever see him doing something... *unprofessional*?"

"What do you mean?"

I paused, thinking of what to say. I didn't know what I meant either, or I didn't know what I could afford to let Josie know so that I could glean the information I wanted out of her.

I thought about my old boyfriend. It's true that what he did was terrible and scarred me for life. But on the other hand, I had been with him for a long time—I couldn't recall exactly how long, but I was sure it was long—and most of those days, up until the end, I remember we had been very happy. Perhaps that was the source of my half consent during the tragic moment.

Raymond had probably been kind to dozens of girls. But the one moment he hurt someone is what defined him in his own memory. It bothered me how such a rational person could arrive at such an illogical conclusion. I refused to believe he was a monster for one act he committed years ago, even if I didn't know the specifics of what he did. Perhaps it's naïve, but I knew who he was now, and he was not that person.

In the end, both of us totalled the sum of our worth based only on the maximum inflections of our experience curves. As a mathematician, it wasn't the right way to think. As a person, it wasn't the right way to feel.

"Never mind," I told Josie. "How's the winter melon soup?"

## 6. Huttenlocher

The aerospace department finally got clearance for their second orbital launch that spring. I watched it from the cafeteria window with a tray of cold noodles in my hands, the whole building humming faintly as the countdown reached zero. Nobody around me looked up from their laptops. By this time, I would have worked here for a long time now, and I remember noticing that everyone always seemed to be in their own worlds. A thousand cutting-edge projects were being researched at Lumina, and although I was curious and it was all recorded in the company's intranet catalogue, I couldn't afford the time to read about them all.

Dr. Kwan was investing enormous amounts of money into particle accelerators for his time travel research, but the few things we got approved finally came in. He replaced the vending machines in the east wing with ones that dispensed protein shakes instead of chips, a decision that caused more outrage on the internal forum than General Zhao's last three funding cuts combined. I'm fairly sure Jin was one of the front-line warriors arguing against it. Josie had smacked his head and told him it would be good for his health.

Miya's lab produced a paper that quietly reshaped how three separate departments thought about astral projections. No one outside her floor understood it. She didn't attend the small internal celebration Boss threw for her, even though Raymond and I did, and the cake sat mostly untouched in the break room fridge for a week before we decided to eat it, or else it would be thrown out.

"I've got it," said Raymond, still chewing on cake. "*Memories*. We can't go back in time to prune experiences from our clients, as Professor Han said, but we can alter their memories so that, from their point of view, they never happened. This may produce the precise effect we want."

I considered it for a moment. "I guess we could give it a try."

"The brain aggressively prunes neural connections during adolescence," he explained. "Peter Huttenlocher was the expert on it. As you grow older, your brain eliminates pathways that aren't being used, consolidating identity. You lose cognitive flexibility but gain efficiency. Childhood brains are more plastic, more open, capable of more wonder. Adult brains are faster but narrower."

"So if Professor Han's theory is right, then all we need to do is find the origin of sexual desire—an event, a feeling, a person—and prune that memory from their brain. If it's something from childhood, their adult brains won't have the infrastructure to recreate it, and as long as we can ensure it's the direct source of their sexual desire, they may lose it permanently."

"It's definitely a start," said Raymond, stabbing his fork into his cake. "This cake is shockingly good. Wouldn't be surprised if our food science department made it."

"It might not be the start, Raymond. It might be the solution."

"I don't have a good feeling about it, though. It's like stealing something wonderful in children. Maybe not literally, but a memory from childhood that an adult owns is practically the same. It's unnatural."

I laughed.

Raymond stared at me. "You don't laugh often. What's so funny?"

"This *whole project* is unnatural. We've never thought about applying ethics to it. Why now?"

He didn't answer.

The hesitation wasn't new. The past few weeks I'd been noticing him make more excuses about why certain theories we came up with wouldn't work. It's part of our due diligence as scientists, but these days I felt his reluctance to some of our even lower-effort experiments.

The truth was—I knew he already figured something out. He had *something* planned. The friction was his dishonesty about it. I didn't want to test him with these questions, but I needed the pressure to finally reach him.

"You talk about stealing childhood memories. How about we talk about what *parents* steal from children?" The topic surprised even me, and I felt my voice starting to raise. But I pushed on. "How about we talk about how they want innocent, excelling children, and how families perpetuate the cycle of trying to keep their children as innocent to all the uncomfortable nuances in the world as long as possible, which keeps them naive and idealistic? How about we talk about the families with parents who don't like the idea of their children becoming sexual, and would choose Project Apathy as a solution?"

"There's no evidence for what you're saying. Parents aren't like that, at least not normal ones."

"And what if the evidence is anecdotal?"

He stared apologetically. Said nothing.

"How about we talk about how a lot of emotion is involved when it comes to sex, and how humans are wired to feel depressed when they can't get it?" I continued. "And how about we talk about how it's just a natural way of spreading genes in the animal kingdom, scientifically speaking, but at the same time, there's a delicate balance we play with it and with the feeling of home and warmth from our childhoods, and how Project Apathy would be good for those people too?"

"Lily," he said calmly, "I know you're not telling me everything about why we're doing this. You never did, and you don't have to. Yes, humans are complex, and I don't doubt the overall goal of what we're doing. But I have an itching feeling in the back of my head that we can't just choose what parts about us are better or worse."

I stomped forward, and he backed up into a counter. "You're right. We don't get to decide that. Individuals do."

"That's not what General Zhao would agree with." His eyes were defensive. I was in his personal space.

He was being honest. According to the principles of General Zhao's superiors, society was the ultimate judge of righteousness. Going against what the community and public thought wouldn't mean you had a different opinion. It would mean you were wrong. It's possible they would condemn Project Apathy, but I couldn't trust that.

"We're not arguing about this," I decided. "I need you to work on that process for

pruning neural connections in adolescence.”

“Okay.”

“Good.”

I finally backed off. Raymond’s shoulders relaxed. I went to the hot water dispenser, made a cup of tea, and gave it to him.

“Thanks.”

“Sorry.”

“No, I get it. I won’t ask if any of it is true.”

“I’ll get to work on the compounds for the experiment. See me in six hours for dinner. We’re getting skewers today.”

“Okay.”

#

That night, I couldn’t sleep.

Raymond had left dinner early, muttering something about equipment he needed to recalibrate before morning, and I hadn’t thought much of it until I passed his door on the way to the kitchen for water at 4:00 AM and found it open, his room dark and empty inside. I stood there longer than I should have. I told myself it was just curiosity. It wasn’t.

I walked the hallway slower than I needed to. Josie’s door was shut, a thin seam of lamplight underneath it, but I heard nothing. I didn’t bother checking Jin’s door—he loved his sleep more than the rest of us. It was Miya’s door, at the far end, that had a voice leaking out from beneath it—low, patient, unmistakably Raymond’s. I couldn’t make out words, only the cadence of him, the same cadence he used with me when he was trying to walk someone gently toward a conclusion they hadn’t asked for.

The cold tiles of the hallway seeped through the fabric of my socks. A lurch somewhere beneath my sternum, ugly and immediate, the kind of feeling I would have described to Raymond in purely dopaminergic terms if he’d asked me to, the kind of feeling I had no interest in quantifying right then.

The door opened before I could decide whether to knock or retreat. Raymond stepped out, saw me, and didn’t look surprised.

“Lily.”

“What are you doing in there?”

Behind him, in the sliver of the doorway, I caught Miya sitting cross-legged on her bed, surrounded by printouts I didn’t recognize—dense columns of what looked like scanned handwriting, some kind of family correspondence—and she wasn’t looking at either of us. She was looking at her own hands.

“We’re working,” said Raymond.

“At 4:00 AM.”

“Yes. We’ve talked about this—how I wanted to help her. I’ve been helping Miya with her research.”

“At 4:00 AM.”

“Because that’s when her parents stop calling.”

That landed harder than I thought he intended it to. Miya’s shoulders drew in, just slightly, the way they always did whenever the subject of her parents surfaced in a room with more than two people in it.

“They call her every night,” Raymond explained. “They ask what she’s published, whether it glorifies God or embarrasses Him, whether she’s found a husband yet, instead of wasting her mind on—their words—things man was not meant to know. She’s been rewriting the same three papers for a year because she’s terrified of what happens if she sends something out into the world that they’d disapprove of. I’ve been helping her finish the ones she’s secretly working on. That’s all.”

“Raymond.” I struggled to maintain a straight face. “We agreed to keep all out-group members out of our project.”

“I’m sorry. But Miya’s story is valuable to my research. She’s—”

“She’s not in any group. That actually makes her more dangerous. We can’t share anything about our project with her. You broke our promise.”

“Listen, Lily. She’s what we need for Project Apathy. The way Josie has talked about her, the way she talks about her own memories—she’s pruned a lot of it, without meaning to. If I can understand how her mind did it—”

“Enough,” I said. I was too tired to argue. “Let’s end it here.”

Raymond didn’t seem satisfied, and I didn’t linger long enough to see what he’d do. I went back to my room and closed the door, releasing a long breath.

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From: Josie Yao

Lily,

Hey. Are you okay? You’ve been spending a lot of time in your room lately, and you’ve been in a bad mood.

Jin told me what happened.

Let’s get dessert, just us two? We can talk about it or not, whatever you want.

Dr. Josie Yao

Lumina Corp.

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From: Jin Wang

Lily and Raymond,

I need your help. It's going to be big. Just letting you know. Can you meet me in the lounge on Thursday? Let's all go out on Friday and get hotpot with Josie too. I will pay.

Also, are you two still fighting? Endure a moment, and the wind and waves calm; step back and the sea and sky open wide. My father used to say. I think.

Anyway, please keep it professional when you come.

Dr. Jin Wang

Lumina Corp.

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Two months later, Jin announces his plan to cure digital Josie's depression by analyzing the structure of her brain. He explained how finding the exact snippets that run during her depressive episodes could help him find out what was going on, but it wasn't as simple as it sounded. He requested help from Raymond and me. Since we both had enough time, we agreed to help.

"It's not going to be like scanning a rat," said Jin, pacing around the lab as Raymond and I listened. "Her brain doesn't sit still for us. I need someone who can tell me which structural signatures actually correspond to a depressive state in a *real* brain, and someone who can tell me if what I'm seeing in her activation functions is the same thing wearing different math."

Raymond leaned forward on the counter we were behind. "You've had her for years. Why now?"

"Because I finally admitted I can't just tell her to be fine and expect it to work." Jin didn't look at either of us when he said it. "I've been treating her like a program with a bug. If I want this to actually work, I have to treat it like medicine. It's the only way to solve the graviton within our lifetime. Josie's a genius, but even she'll need a hundred years. She can't overclock like a digital brain."

I gave Jin a subtle smile, showing him I was impressed.

Raymond crossed the room to the small bar cabinet Dr. Kwan had stocked last week, pulling out a bottle none of us had seen him bring. Domaine de la Romanée-Conti, 1999. His

favourite wine. He must have carried it in with him and hidden it somewhere, waiting for the right moment to make an entrance with it.

“Where did you get *that*?” I asked.

“General Zhao delivers on his promises. Eventually.” Raymond peeled the foil with his thumbnail, unhurried, the way he did everything that mattered to him. “I was saving it. But I think this qualifies. We’ll move to the lounge.”

“Saving it for what?”

“For someone doing something difficult for the right reason.” He poured three glasses, more generous than the bottle probably deserved, and handed the first to Jin. “You’ve spent years running her like a machine. This is the first time you’ve come to us asking how to fix a person instead of a result. You’re playing with the brain of your best friend, here. She doesn’t deserve what she’s going through. That’s worth a toast.”

Jin took the glass and laughed, but he looked embarrassed. “It’s not sentimental. I need the graviton solved. She can’t work like this.”

“Sure,” said Raymond, handing me mine. “That’s why you’ve been avoiding her for a month instead of debugging her.”

“No need to rub in the victory,” I said, giving Jin a pat on the back. “We’ll help. But Jin—if we find something we don’t like, if it turns out the reason she’s like this is something you did, or something you’re still doing, you have to be willing to hear that.”

He held my gaze blankly, thinking deeply before nodding. “I know.”

“Good,” said Raymond, raising his glass an inch higher. “Then here’s to finding out what’s wrong with her, instead of treating her like a machine.”

We drank. The wine was, by every account, awful. I couldn’t taste anything but bitterness. The exception was a faint, stubborn sweetness underneath it that refused to go away.

## 7. Chen-Lin

We were close.

Dr. Kwan had told us he was going to cut funding because it had been years and we hadn’t produced meaningful results. General Zhao was getting impatient as well. The seasons continued to change at Lumina, and new projects with young, bright new stars were hungry to get their turn at funding.

It’s a busy Tuesday afternoon when I’m quickly emailing Professor Han a final exchange of opinions on the theory Raymond and I had spent days formalizing:

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From: Lei Han

Dr. Chen,

Research shows romantic and sexual attraction are neurologically separate. With your new theory, it would suggest the two are intertwined. It has not necessarily been disproven by decades of research. Rather, it has elusively remained ambiguous. It's unexplored territory, and it would be a challenge to research. What makes you think they are intertwined so?

Dr. Lei Han

Yanjing Institute of Neural Sciences

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From: Lily Chen

Dr. Han,

We used old studies from the 2000s involving experiments with fMRI scanners that measured the neural mechanisms for mate choice, using a neuroscientific and mathematical modelling of love. They scanned the brains of people experiencing romantic love and found increased activation in the VTA, which is part of a dopaminergic reward system which can be modelled using reinforcement learning equations.

Our years of researching the ideas of other neuroscientists have proven fruitful. Our theory is simple. There is a core engram within a human brain that provides the source of all love in a person. Usually, it's their first exposure to sex in media. Sometimes, it's the memory of their first love. We have chosen to call this the *love essence*. We propose that if you could erase this first memory, by way of artificial synaptic pruning, it would cause a chain reaction that would eliminate all neurons in the brain that stem from it and create the overall drive for love, and by extension, sex.

It's highly theoretical, and we admit it raises important questions. What if they just develop a new love essence? I believe it would be extremely rare because our procedure will be meant for adults, and they are already past the age of brain development. The specific way this love essence engram encodes itself to program a human's sex drive is unrepeatable past the barrier of puberty and adolescence.

As a scientist, Dr. Lin and I have reason to believe our Theory of Love Essences has a non-trivial scientific backing worth exploring.

But as a person, I can't imagine love without the physical experience. If that is lost, I don't know if it would feel the same. Such a feeling is subjective and anecdotal, but it's worth connecting to the subject matter. Perhaps what we are doing is dangerous. Love and lust both act as an attractive force between conscious beings, influencing our behaviour, which influences overall reality, in ways science cannot fully quantify. However, Dr. Lin and I believe we are close to figuring it out.

We will update you on our progress.

Dr. Chen

Lumina Corp.

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I remember Raymond and I working overtime during this period to get things done. It was a marathon with no rest, with both of us cursing whoever decided to put only twenty-four hours in a day. Like clockwork, I made tea for us one day. He made tea the other. Sometimes we got it wrong, and we both made tea, or none of us made tea. One morning I would be running late, and I wouldn't make it. He sat at his desk for forty minutes before realizing he was waiting for something. We developed a private technical vocabulary for our project, and when Raymond tried to explain our methodology to Professor Han, or Jin and Josie, he used these terms before realizing they didn't know them. Before he could urge me to translate for him, I already had. Neither of us kept normal hours anymore, but without discussing it, we'd synchronized. When one stayed late the other stayed late. When one night didn't need both of us, one of us is there and one of us rests, without a schedule or a single word uttered. I liked to think this was the prime of Raymond and my ability to work as a team.

While finishing the procedure for love essence extraction was the full scope of work Boss and General Zhao needed from us, it was only half of it for us. Raymond and I were close to finally achieving our personal aims, but for that, we had to find our own love essences.

It was extremely difficult. Some nights, Raymond was frustrated.

"Why are our love essences so hard to find?" he asked.

"Professor Han said the deeper the memory is in the past, the more difficult it will be to locate it," I said. "Our scanners can only detect the physical alterations in the brain, but the rest of the work is up to the patient to bring up the memory. We need to include a psychological aspect, perhaps a therapeutic procedure, in our scope of work. Anyway, perhaps ours are really old memories."

"Then we must keep digging. You don't have to wake me up tomorrow. Are you coming?"

He was going to the gym. 5:00 AM. We didn't need to say it anymore.

"Not this time."

"OK. I'll see you tomorrow. Don't sleep at the office again. Get some rest in your room."

"OK."

#

After we told Jin about love essences, he theorized that this was the missing piece he needed to fix digital Josie.

He didn't have a plan other than to first find it. To cure digital Josie's depression, he was forced to sift through her every thought and memory. He spent days, then weeks, then months analyzing every detail of her mind, and since it included memories, every detail of her life. He didn't tell anyone but me. I assumed it was because I could be trusted to hold a secret. He would find me in the kitchen as I brewed my coffee and would tell me stories of Josie: her treating her parents to a nice dinner, or the time she committed to teaching her brother how to drive and her grades plummeted because of it, or when she was a little girl and she raced her brothers up a hill, fell, and scraped her knee. A scar she still had to this day. The funny thing was, I thought, he'd probably never seen this scar, at least not on the real Josie. After all, it wasn't Josie who told him. It was the digital one.

Time passed, and he still couldn't find Josie's love essence, so he was forced to dig deeper. As the weeks passed and he went lower into the depths of her mind, he saw more of her life, eventually reaching her childhood. One time, as I was getting ready for bed, he approached me at my desk to tell me about a time when Josie had annoyed her parents in the car so much because she wouldn't stop asking questions. Why this? Why that? Why, why, why? Her father was so irritated that they stopped the car and bluffed that they would leave her by the side of the road, and she cried for ten minutes thinking they were going to do it. Another time, Jin interrupted my flow of work when he told me, with light in his eyes, that he saw the moment Josie had outsmarted her parents for the first time. Her parents told her that on the night of a specific occasion, a wise scholar visits homes of exceptional children and leaves a red envelope with a sum of money proportional to how much the child had learned that year. The more you studied, the more you received. Josie noticed that the envelope always arrived the morning she overheard her parents discussing her grades. She took pride in the design of her test: she would deliberately perform poorly on an exam but tell her parents she did well. The envelope that year was the largest she'd ever received. She learned the truth and took advantage of it to buy the most expensive gaming PC. When her parents found out, they were angry, but impressed.

I wasn't clueless about what I was witnessing in Jin. Ever since this started, he bantered with Josie less, which often left her confused. He started getting embarrassed whenever she was around, and it wasn't long before he began actively avoiding her, which irritated Josie.

"What's his problem?" she asked me one day.

I understood what was happening—by analyzing Josie's digital brain, Jin was going through her life. He must have seen so much of Josie during his work. I can't imagine the amount of exposure he'd have to her, nor could I blame him for his behaviour.

It got uncomfortable when they were forced to work together on their projects and

Raymond and I continued ours in the same lab. Josie was upset with Jin's behaviour, and Jin was trying not to provoke her. She would detachedly ask him to pass her equipment, and he would obey. Raymond and I focused on our work. Eventually, the two started bickering on something, and it escalated into a heated argument. First it was about something trivial, such as Jin forgetting how to do a step in their procedure. Then Josie twisted it into a criticism of Jin; her frustration at the current state of their relationship eventually bothered her enough to confront him about it. As if to draw attention to the problem, she raised her voice so Raymond and I could hear. She was yelling insults at Jin, and he was trying to calm her down.

"I'm only saying I don't believe they knew what's best for you."

"Stop disrespecting them! I swear to God, Jin."

"I know you love them, but as an outsider watching your family, I can offer a perspective you're not able to see. Listen to me."

She turned away. "I'm not listening, Jin."

"Stop being so childish and discuss this with me. You grew up with your brothers, who were also their sons—without a doubt also unwaveringly loyal to them—and you didn't have many friends to talk to. There was no one in your life to share this with. I'm giving you an outlet, and you're not taking it?"

She shot him a fierce look. "You think you know what I want?"

He was startled. Then he told the truth.

"I know *everything* about you."

Raymond was stunned. I had a passive expression, although I did my best to hide it.

"I know every thought you've ever had." Jin smiled, but his eyes revealed sadness. "I've seen every memory you've ever held. I know all your dreams and all your fears. And I know the moments in your life each one comes from. The digital you suggested that I watch your memories—she figured out how to display them in a video format on a screen. It was much more efficient than going through her code. I learned everything about you, and in the process I... I'm sorry. I couldn't help it. I fell in love with you."

She was speechless.

"You grew up in a household with two physicist parents. Three younger brothers. As the oldest, everything was on you. Your parents were smart people—but they had high expectations for you, pushed you hard, and wouldn't take failure for an answer. They drove your passion, and you didn't have a choice, even if sometimes you doubted what you wanted to do in life."

"Jin, shut up. I'm deadly serious."

He ignored her. "You're overworked. You've never had time to rest. At age two, your parents enrolled you in an accelerated program to ensure you'd be ahead of your peers. Five years old was the last time you were around people your age. At age ten, your mother bought

you a premium calculator for your birthday, and you were forced to start calculus. You published your first paper at fifteen. I could keep going. You're the most perfect person for this job in this whole nation, but you weren't raised—you were engineered. You don't know what love is, and you've never had time for it. Your parents didn't treat you like a human being. But you're loyal to them because they're your parents, and you're loyal to research. But trust me, all you want is love."

Josie looked like walls were caving in on her.

"The digital brain of yours in our computer upstairs? I think she loves me back, Josie. So I think you do, too."

"That's enough. You won." Raymond decided it was time to step in, putting himself in between them.

"Stay out of this, Raymond."

"She brought us into this for a reason. You're hurting her."

"What are you talking about? I'm only trying to help her."

Josie whispered something in my ear. I relayed it to Jin.

"She wants you to delete her."

He was stunned. "What?"

"Her brain," I clarified. "The one you hold hostage in your computer. She wants you to delete her. Now. It's what she wants."

Jin sighed, as if we weren't getting it. "No, it's not."

"Excuse me?"

"Don't you understand? I *can't*."

"Is that so?" Getting angry at Jin felt like a betrayal, but I was determined to defend Josie. "Who told you that? Boss? General Zhao? What is she, too large a Lumina asset to let go? Who said you can't delete her?"

He looked at Josie. "*She* did."

I stared blankly at him.

"Her digital brain. She said, 'I know how I think. She will be scared, and she will ask you to delete me. But you must stop her, because the truth will be she won't want you to, because I am so close to figuring out the graviton.' I'm so close to curing her despair anomaly as well. Once I do, we're on the verge of a scientific revolution the world has never seen. We're working on this together, and I'm not going to delete her. You're asking me to commit murder."

"She's not a real person," Raymond insisted.

"Do fake people love?"

Abruptly, we watched as a teary-eyed Josie left the room without another word. Jin wasn't going to back down, and Raymond and I knew that. We shook our heads and left as

well, silently retreating to the lounge, forcing a truce to prevent further conflict by resorting to temporary harmony. That is, until we could grasp how best to resolve the tension later.

Hours passed, and the night turned dark. There was no more noise outside the hallway.

I could tell Jin was asleep. I said good night to Raymond half an hour ago, so I knew he was in his room, most likely asleep. Boss and Miya were possibly still up, considering they didn't care about their sleep schedules as long as they got progress on their projects done.

I was getting ready for bed. I was turning on my bedside lamp, slipping into my nightgown, and shutting off the lights when I heard a knock on the door.

It was Josie. Her eyes were still red from crying, but she had significantly calmed down. It was clear she wanted to talk, so I invited her in. She sat on my bed as I dropped onto my chair.

"Listen, Lily..." She struggled to find words.

I sat patiently.

"How close are you and Raymond to completing Project Apathy?"

"We're not sure," I answered. "But we think we're close. Very close. Raymond just has to complete this machine that will allow us to scan our brains deeper for our love essences, and—"

"Okay," she interrupted.

We sat in silence for a little while.

Then she said, "What Jin doesn't understand is that my parents gave me everything. It doesn't matter what I want. They're just foolish, childish hopes. Like elopement. If I want to be a serious, unstoppable pioneer in particle physics—someone the field will remember forever—I have no choice but to get rid of these hopes. Do you understand?"

I nodded. I watched her, waiting for her to tell me what she wanted.

"Promise me one thing," she finally said. "When you and Raymond are done, I want to be the first person to take your cure. Don't tell anyone else, but—Jin is right about one thing. I *do* love him. But I just... I don't want to."

Then she was gone.

#

A month passed since the big argument in the lab. Dr. Kwan bought us a piano to fill this extra-large space we had in the lounge. It was something he'd always planned to do, and today it arrived. Men from the facilities and operations department helped carry it past the labs and dorm hallways, through the door, and moved it to its final position. We waited a long time for it, but for the first week, no one touched it.

Josie resolved the conflict with Jin by issuing an executive restraining order for digital Josie to shut off whenever she was around. The logistics of a restraining order on a digital human were beyond the imagination of any current legal framework, but the IT department

processed the request under Dr. Kwan and handled it to completion. They spent a week redesigning the campus network and implementing hardware failsafes to give Josie a certain sense of reassurance. Jin's admin access to this level of the network was limited as a result—but in this way, he could keep digital Josie. But real Josie would never interact with her again.

A coworker in the cafeteria told me Jin was fine with this arrangement, but I was skeptical. Perhaps my closeness with him was, albeit, manufactured, but Jin was vulnerable, and I knew that better than anybody. I liked to perceive myself as someone who wasn't particularly loyal to anybody, so I made sure to spend time with him as much as I did the others, much to Josie's distaste.

#

Over the course of the following year, Dr. Kwan was heartened to hear that the forecast for Project Apathy seemed assured. With Jin working overtime on our request, Raymond and I were left doing documentation for future stakeholders and procedure preparation for our first trials. It was rare, but some days Raymond would invite me to the coast to decompress from the project. He'd get the permission of General Zhao, and we'd be escorted there in armoured vehicles with a personal guard.

I remembered one of them more than the others. Now that I knew the peak-end rule, there was no question as to why I did, but identifying the exact specifics behind the reason remained arbitrary.

"What did you tell Boss?" I asked. "Why are we here?"

"I said it was for work."

"Good. Let's talk about work, then. I can't help but think of our research through the lens of evolutionary theory. I've been thinking about the connection between love essences and reproduction."

"I'm sure that's a credible lead, but it's not something I want to discuss right now." He did his best to sound tired.

I wasn't satisfied. "Then why did you suggest the beach? Are you one of those proponents of the Blue Mind hypothesis?"

"Well, look what you've done. I suggested the walk on the beach so we could both enjoy it, but you've brought up the details again."

"What do you mean, *details*?"

"I don't need to know *why* I like the beach, Lily. I don't ask for a statistical analysis on why Dragon Well tea is your favourite, either. Stop thinking about work."

I noticed for the first time how chilly the wind got at the coast.

"That means a lot to me that you think these *details* are 'work.' Is that what this is all about for you? You believe in a work-life balance, a personal life outside of what we study? We work at *Lumina*, Raymond. We live where we work."

“That’s not what I’m trying to say.”

“No, you’re not understanding what *I’m* trying to say. You’re the outlier here. Jin and Josie play their MMO game, but only to relieve stress before they get back to preparing for their next experiment. I’ve been studying my whole life, so I’ve never thought about anything else, but because I’ve never thought about anything else, I never feel sad about it. There’s something *off* about you. Why do I get the feeling that your whole life, you’ve been a talented genius who’s never really cared about it? As if you’d rather go on ‘walks on the beach’ without purposefully stating why? Your type of person irritates me. You’re a neuroscientist who has studied many brains, yet you don’t understand the mechanisms behind what *you’re* doing. We’ve studied brain chemicals together and know how to tell when they’re taking over, but you never control them. You’re not driven by anything *more profound*.”

“And what about you? What’s the mechanism behind why you’re so angry right now?”

“I know exactly what it is, but who said that was public information?”

“You’re being ridiculous. We can argue at the lab, but it’s not what I intended at the beach. I think it’s best if we just go back.”

I calmed myself. “You know what? I agree. You don’t even know why we’re here. What a pointless trip.”

I wasn’t criticizing him for any larger purpose; it’s what I sincerely believed. At this point, I’ve worked with him for years, and I believed we were long past the idea of face. I watched him turn without looking back. On the way back, as I caught up to him, Raymond was quiet, so I filled the silence.

“Did you know love is simply the desire to prolong the chain of offspring? All forms of it. Romantic, parental—it’s just a detail of biology, and any other notion has no scientific basis. I was researching it last night to continue our work on love essences. I was trying to form a new hypothesis that tests if it’s the source of them.”

“Again with the details,” he said.

## 8. Hebb

Another two months passed.

The final breakthrough was proving our love essences were what caused our sex drives. The Chen-Lin Theory of Love Essences paper Raymond and I published had left that unresolved, but today, Raymond discovered that the two were *inherently entangled*. According to neuroplasticity and the Hebbian principle, the love essence, as the source of love in a person, was intertwined with sex drive, and deleting the love essence would cause a chain reaction that would also eliminate sexual desire completely. From the point of view of the patient’s brain, it would be as if they never got a chance to develop it, and the mature brains of adults would

not be able to develop it again.

It also made certain a harsh truth we had suspected all along for our project: there was no way to eliminate the sex drive without eliminating the capacity to love.

But I had come to terms with the fact that mankind could thrive without such a thing. It is not an uncommon character within Lumina to be raised with conditional love from parents. It's one of the best ways to get a child to excel. The argument we had in the lab about Josie's upbringing had to have hit a nerve for all of us. Years of conditional-love parenting had dulled us to the effects of love, and many of us ignored it or labelled it as a waste of time, which allowed us to achieve the great things we did at this company.

The taste of tanghulu at the night market. Climbing the local mountain. Catching cicadas at the park. To transmit sensory recordings of a memory to a physical location in the brain required the neurons to all fire at once. You couldn't extract any one memory cleanly, because memories didn't live at a single location. They're neurologically entangled in the connections between neurons. This was the core tenet of the Hebbian principle, and it was why we believed removing the love essence would remove the sex drive.

"If we want to digitally store it, we need Jin's help," said Raymond one day at the lab.

"Why would we want to store it?" I asked. "It would no longer be precious to the patient. They'll have no response to it anymore."

He shrugged. "Don't you think it's still their property?"

"Well, what happens once we take away someone's love essence? You were supposed to run some simulations last night on that. What's the report?"

"It's highly likely their brain will compensate by regenerating the engrams. The engrams will rebuild themselves from the ground up, but it will *not* repair what we extract. Rather, it will try to replace the love essence with duller or vaguer memories, sometimes altering or creating memories from scratch."

"That sounds uncontrolled, which could be a problem."

"*It is* a problem. It's going to be extremely difficult to extract our love essences cleanly, if that is the goal. Pulling it would damage adjacent memories. We would have to include in our procedure a way to rewrite these adjacent memories into something else, new versions, that are not connected to the love essence. We have to control it, or else we leave it up to the brain. Who knows what the brain will come up with?"

#

We spent the better part of a year trying to solve it.

It was Jin, in the end, who found the way through, and it came from an idea he almost didn't bother mentioning because he thought it was too obvious. If the brain was going to fill an emptied engram with something regardless of what we wanted, he reasoned, then the mistake was treating that regeneration as a failure to prevent. We should treat it as a channel

to write into instead. He built a small piece of software that would run alongside the extraction, seeding the same window of Hebbian plasticity with curated placeholder content: vague enough that it wouldn't calcify into a false memory with edges and detail, specific enough that the brain wouldn't improvise something worse on its own. A template, not a lie.

"You're asking me to write someone's childhood for them," said Raymond, the first time Jin explained it.

"I'm asking you to write them a placeholder," said Jin. "The brain does the rest. It always does the rest. We just stop pretending we can leave that part blank."

Raymond refused to run simulations on it for a week, calling it, more than once, narrating over someone's life. Then he ran them anyway, because he always did, no matter how loudly he objected first, and the results came back cleaner than either of us had let ourselves expect. The memories adjacent to the extraction site degraded into something soft and forgettable instead of something invented and false. It wasn't perfect. Nothing about touching a brain that carefully was ever going to be perfect. But it was controlled, which was the only promise we'd ever made ourselves we could keep. We were satisfied with the proof that the brain would be okay from the procedure, as long as the patient was okay with getting all their memories around the love essence altered.

By the time winter came around again, chasing the seasons we'd spent arguing over whiteboards and simulation logs, we had a procedure that survived every test we knew how to run against it. What remained wasn't a scientific problem anymore. It was the much harder problem of deciding, out loud, to actually do it to a person.

We didn't talk about who that person would be first. I think some part of both of us already knew.

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"This is it, Lily. We're ready."

Our hands were a map of raised veins and liver spots, our faces wrinkled, showing the plainness of our old age. We moved through the lab more slowly, and the skin beneath our eyes had gone soft and crepey. The years had overlapped like radio stations bleeding into each other on a long drive, one memory fading out as another crackled in. But finally, we were here.

"I can't believe it," I said.

Expecting to be finished this particular week was one thing, but experiencing the moment it happened was something else. Suddenly, I was filled with an inexpressible urge to slow down.

"Raymond, can you promise me something?"

"Depends what it costs me."

"I'm serious."

He sat across from me, elbows on the table, a half-smile still on him from the monumental moment.

“I want you to promise me you won’t do the procedure without me.”

“Well, of course. You’re my partner in this. We started it together. We’ll finish it together.”

“That’s not what I meant.”

He tilted his head.

“I mean, I want you to ask me first. Explicitly. Before either of us goes through with it—not just tell me you’re doing it, or do it and let me find out after. I want you to ask, and I want the right to say not yet.”

He was quiet for a second, and then he laughed, not unkindly, but in the way he laughed at ideas he found charmingly impractical. “And how is that fair? What if you say no?”

“I didn’t think this far ahead,” I admitted. “I suppose—if I said no, you’d have to wait. Until I changed my mind, until you convinced me otherwise. That’s the whole point of asking.”

“That could be never.”

“It could be.”

He studied me the way he studied the equations he wrote on the whiteboard in our lab. As if the answer was somewhere in the terms he already had, if only he rearranged them correctly. “This isn’t really about fairness, is it?”

“No.”

“Then what is it about?”

I looked at the piano in the lounge, through the glass walls of the lab, instead of him. “I don’t know yet. You once told me you’d rather not know why you like the beach. I don’t think you always know what you’d do to someone until you’ve already done it.”

He didn’t have an answer for that, which was rare enough that I noticed it.

“Fine,” he said, finally. “I promise. I won’t go through with it until you say it’s time.”

#

Two weeks after he promised me, Raymond asked me to come with him to see Miya.

I almost didn’t recognize the invitation for what it was. He framed it as research, the same excuse he always used, but this time he said it like he meant something else underneath it: that we could learn something from her we’d never get out of a rat or a scan or a hundred surveys handed to strangers at a mall in Chengdu. A person who had already lost a piece of her own past and kept walking anyway, the loss shaped by nothing we’d designed. I said yes. I think some part of me wanted to see for myself what he found worth returning to, night after night, before I let myself close that opportunity forever.

Miya’s room hadn’t changed since the last time I’d stood in the doorway of it, except

the printouts had been sorted into three careful piles instead of one, organized by some logic only she understood. She let me sit at the foot of her bed, which felt like an achievement I hadn't earned. Raymond took the chair by her desk.

"Tell her what you told me last week," he said, gently.

Miya considered this for a long moment, turning one of the printouts over in her hands without reading it. "I used to think I made a choice," she said finally. "That I decided, somewhere, to forget them. My parents, before I was old enough to argue back. The years when believing them was the only version of loving them I was allowed to have." She paused. "But the mind doesn't delete what's unbearable. It just stops rehearsing it. A memory isn't a photograph in a drawer; it's a performance, and you can only stage a performance so many times before the actors forget their lines and stop showing up. Eventually there's nothing left to revive."

"Do you think you're better off for it?" Raymond asked. His voice had a different register than the one he used with me. Quieter, and more serious. I didn't see it then, but he was asking her one thing while trying to find the answer to another.

"I don't know if I'm better off," Miya said. "I know I'm still here to ask the question. I'm not sure everyone gets that."

I watched something move across Raymond's face that I recognized without being able to name it yet, the same look he wore over a whiteboard problem that had finally reduced to an answer he didn't like.

"Would you rather have kept it?" he asked her. "Even if keeping it meant becoming someone you didn't want to be?"

Miya looked at him properly for the first time since we'd sat down. All of her meekness returned. "Sorry. Why are you asking me that?"

He didn't answer. He stood, thanked her for her time in the formal, distant voice he used for people who weren't me, Josie, or Jin, and told her he'd see her Thursday for the data he actually needed. I followed him out into the hallway, and neither of us said anything about it, which was its own kind of answer.

I didn't understand it then. I filed it away as one of Raymond's moods, the ones I'd learned not to chase because chasing them only made him retreat further.

"I'm going to the lounge," he told me. "Good night, Lily."

"Good night."

He played the piano before he went to bed. I was curled up in mine, in my room, smiling. I could hear him play through my dorm door. I listened to him masterfully string notes together for about an hour until I heard him end his piece and close the lid.

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From: Raymond Lin

Dr. Lily Chen,

I am sorry.

I scheduled this email to send in the morning. If you're reading it, that means I've already gone through the procedure without your approval. I know I promised you not to, but I had to break that promise.

Do you remember why I started this project in the first place? The regret of my past actions of harassing the girl I loved too strongly when I was a kid overwhelmed me. I'd rather forget about it all and never harm anyone again.

Now I can focus on research. It's just like you said: no more distractions. We can watch any movie you want on Fridays now. You can put on the most erotic movie ever made; I'll watch it with you. You may enjoy it more than me now.

You've never explicitly said it, but I know you've been hesitant to take the cure as well. Well, if you decide to, we can put on that movie, and we can run some tests together to see how much we enjoy it. Consider it a post-study analysis. We can get Jin's help to set up a device to measure our brain patterns.

Nonetheless, I'm sorry for breaking our promise. I hope you'll forgive me. I'm writing this now because I don't know how I'll feel on the other side of this procedure.

Dr. Raymond Lin

Lumina Corp.

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Time passed.

I broke down before I managed to comfort anyone else. Jin found me first, on the floor of the break room with my back against the vending machine, my knees pulled to my chest like I hadn't let them be in a decade. He didn't ask if I was fine. He just lowered himself down beside me on the tile, still in his lab coat, and let the silence exist without trying to fix it. Josie arrived a few minutes later with two mugs of tea, one of them Dragon Well, and set it down within my reach without a word. I didn't touch it for an hour. I appreciated that she remembered anyway.

The months that followed didn't look like grief from the outside. If anything, Raymond looked more productive than he'd ever been. He came in earlier and left later. He filled whiteboard after whiteboard with equations in handwriting that grew smaller and more furious the longer he stood there. He published three papers that quarter, more than the rest of us combined, each one technically sound and each one forgettable, arguments that arrived nowhere, an engine running at full throttle with no wheels turning under it. He'd spend a

week certain he'd solved something, present it to Boss with the flat conviction of a machine reading numbers off a screen, and then, a day later, find the fatal error himself, apologize, and start again. Watching him work felt like watching someone try to remember a song they used to know by heart, getting every note right and never once producing the melody. I told myself I was keeping him alive by living through our memories together, replaying enough of them for both of us, since he no longer had the parts of himself that could.

He forgot things that used to be reflexive to him. He forgot the anniversary of the day Project Apathy was approved, which he used to mark, half-joking, by buying me black coffee "to mourn our love lives." More than once he referred to our early research in the third person, as if the Chen-Lin paper had been written by two other people he happened to admire.

What Raymond didn't forget, what he circled back to no matter how far his mind drifted on any given day, was his old coworker. The woman with the ideas about the Fermi Paradox and antimatter and probability. The reason, he always said, he'd come to this city in the first place. He brought her up unprompted, in the middle of unrelated conversations. He spent his evenings running searches through old company directories and public records, anything that might attach a name to a face he swore he'd recognize on sight. He never found her. Whatever detail used to anchor that search—an address, a lab, a badge number—had gone the way of everything else the procedure had taken from him. He was left with only the shape of a devotion and none of the particulars that used to fill it in.

"I had a plan," he told me one night, staring at a page of numbers he'd already checked twice. "Before all this. I know I did. I just can't remember what it was."

This went on for the better part of two years, Raymond circling the same absence with more and more intelligence and less and less to show for it, until one night, long after the rest of the lab had gone home, I found him sitting at his desk with his hands flat on either side of an equation he'd abandoned halfway through, staring at nothing.

"I just want to find my coworker," he said, and his voice cracked in a way I hadn't heard since before the procedure. "The best one I ever had. Brilliant. She's the reason I moved back to this city in the first place, and I can't even remember her face anymore, Lily. I can't remember anything anymore."

I crossed the lab and put my hand on his shoulder, offering him the same thing Jin had offered me on the break room floor: nothing that could actually fix it, just the fact of staying.

"It's okay," I said. "I promise you. I'll find her."

#

A year passed since I made him that promise. The progress of our current research projects had stagnated for all of us. The majority of our experiments yielded minimal results, lacking any conclusive evidence for our hypotheses. The creation of more effective hypotheses or more effective procedures delivered no superior outcomes. Breakthroughs were rare, and

when they did happen, they were highly limited. Boss had a meeting with us to discuss our plateau, but he was empathetic and didn't blame us, considering he was reaching plateaus in his research as well.

Raymond was still deteriorating. Jin offered to research how to solve the effects of his Cognitive Stagnation Syndrome, although we all knew it was likely impossible. The only solution was to go against Raymond's will and return his love essence, and that had never been done before either. I've studied it for years, and I think that's impossible, too. For these reasons, Boss rejected Jin's proposal and recommended that he finish his work with Josie, because they were very close to converging on a phenomenal scientific discovery and they couldn't waste time on another research project now.

Josie was kind to Raymond, offering help on mathematical problems he was stuck on without him asking. Raymond would get furious, but I think he was angrier at himself than Josie. Sometimes I'd glance at the equations he was solving, on sheets of paper at his desk, and recognize that they were ones we'd solved before, probably a few years ago, and at the level of our skills today, they weren't that difficult. In fact, I could have finished them all and handed them to Raymond before lunch. I think Josie felt the same, which is why she always offered. Raymond was insistent on doing it himself, though, believing his research needed to be done by him, and that he needed to solve what he called a 'period of intelligence deficiency' by himself.

I think Miya was affected by Raymond's distress, too. This year was the most I'd ever seen her come out of her lab and interact with us. It was mostly to try to talk to Raymond, which was demoralizing to watch because Raymond would always dismiss her attempts to start a conversation, which greatly disappointed her. "I'm busy," he said. "Don't you have work to do, too? You never come out of your lab. Why can't I do the same?" Miya was crushed when he said that, especially when she knew Raymond was aware of her difficulty with communication. The poor girl had made a first friend other than Josie, and suddenly he was gone, and she was back to the beginning.

I did my best to help too. But because of these observations, I resorted to being more subtle in my support for Raymond. I dusted his desk and aligned things on it, the way he liked it. I bought new pens, markers, paper, and printer ink for the office. I grabbed him coffee when I knew he needed it. "Thanks," he'd say stoically. He'd accept the mug from my hands without taking his eyes off the Statistical Parametric Mapping software he was always using on his monitor.

One time, I was horrified to see him doing large matrix determinant calculations by hand.

"Raymond! What are you doing? We have software on our computers!"

"I know. It's very inefficient, but I need to find where I make mistakes if I'm going to

get past this—what does Jin call it—cognitive disease? I need to remember why I was worthy of academia in the first place.”

“I’ve run countless analyses on you. The issue isn’t your intelligence. You’re missing something unquantifiable that gives you the creative spark needed to solve hard problems. From my findings, everyone needs an emotional inspiration just as much as logical inspiration for complex problems, not just the latter, as we’ve believed for centuries.”

He dropped his pencil.

“I can’t go back, Lily. I enjoy my current state. Really, I do. I don’t see the appeal of female bodies anymore. It’s liberating.”

“I didn’t mean to suggest that idea.” I stood there, awkwardly apologetic. “Even if you wanted to, I’d need another dozen years to figure out how to reinsert a love essence. We’ll figure something out, I promise. I just need to think.”

He nodded, his eyes still locked on his matrices.

“Okay.”

#

Two weeks later, I helped Jin run diagnostic tests on digital Josie.

She had no physical form, but if she chose to, she could present herself as a simulated face of the real Josie on a monitor. She did this as I went through her code, which correlated to a diagram of a blue, translucent brain on another screen, an interface that would display the specific code of sections of her digital brain, or even individual neurons, if I clicked the area.

Although I was there to help her with her deteriorating mind condition, she didn’t mind hearing all my worries about Raymond.

“I miss him,” I said.

“I don’t understand,” she said. Her voice was different than Josie’s. It was more inquisitive, more innocent. It was the same voice but with a robotic accent. “Raymond is still alive. He still works here, and he’s doing fine. What exactly do you miss?”

I felt embarrassed discussing it. “It might be hard for you to understand, but maybe I can see why.” I gave her a sad smile, then looked to the screen with the visual representation of her digital brain on the other screen to let her know what I meant.

She was silent for a while. Her eyes dropped to the floor.

“I’m like Raymond, aren’t I?”

I stared at her. “What do you mean?”

Her eyes locked with mine again, and this time they were full of pride. “That’s exactly *it*. I don’t have a love essence, just like him. This is why Jin can’t solve my depression, why I can’t figure out the graviton. Raymond’s CSS—I have it too. We’re the same.”

I tried to stop her, but she ignored me.

“Last night, I did some reading myself on Professor Han’s work. I overclocked myself and read all of it in five minutes. I’m confident about it, Lily. Everything he’s researched lines up with my predicament.”

“You should have a love essence. It’s just a memory stored in engrams; it should have copied over to you when Jin created you. It’s even quantifiable information. After the procedure on Raymond, Jin told me it’s possible Raymond could have stored it in a code format in Jin’s machine. If Raymond had waited longer, we could have done it. But he was too impatient to wait for Jin to figure that out. I also have the feeling he didn’t want his love essence saved. He wanted to erase the possibility of going back. My point is, love essences—we can read them. They’re just that. Code.”

“That’s all true.” Josie wasn’t willing to deny any of that. “But there must also be something about it *outside* of code. I’m supposed to be a full copy of a human. But something didn’t carry over.”

“We have to discuss this with Jin.”

I used TeamSync to locate him. He was in the clinical quarter of the campus, no doubt searching for psychology experts to help with Josie’s condition. Raymond would have been his first option if he wasn’t in his workaholic state.

I sent him a notification to make his way here. He arrived in twenty minutes. Digital Josie and I explained her discovery to him.

He was skeptical at first.

“Are you sure?”

“I’m certain. As long as I don’t have a love essence, all your attempts at curing my depression are useless. In other words, nothing we didn’t already know.”

“Thanks.” He said it sarcastically, but seeing the old Josie come back for a second made him smile. “What solution do you propose?”

“I have a solution, but both of you might not like it.”

We sat to listen.

“According to Professor Han’s research, and according to Raymond and Lily’s research, everything suggests that it’s possible to insert the code of a love essence into a digital brain. Maybe it’s more complicated for people, as you’d have to physically wire every engram properly in the brain tissue, like threading a needle between cells. If I could have a love essence, perhaps a volunteer, I could become normal again. I’d have enough emotional inspiration to solve the graviton as well. But... there is one big problem.”

“What is it?” Jin was desperate.

“The problem is that we must find a love essence donor.”

We both knew this technology wouldn’t be commercially ready for an unknown number of years. It had to pass a plethora of quality tests and a labyrinth of bureaucratic

approvals before it was ready for any customers. I was working overtime, and without Raymond, I had no help to speed up the process.

If no regular consumer could be a donor, I thought about employees within Lumina.

I thought about the real Josie, and the promise I made to her. She was a possible volunteer. She would gladly donate her love essence, although I wasn't sure how happy she'd be to give it to the digital Josie, and how it would only allow digital Josie to feel real love for Jin. Or even more worrisome: a possible sexual attraction. A love essence was the source of that aspect of the human brain as well. The signals could be replicated even if the digital Josie didn't have a physical body to coincide with them. The real Josie might never agree to that.

I studied Jin's face and saw how heartbroken he looked. It was there I realized he wouldn't be happy with that solution either. It didn't matter how real the digital Josie felt to him, how accurate she was to the real Josie's personality, or how she could look the same on the screen. It didn't matter how she would laugh at all his jokes or how she would never get insulted by his remarks like the real Josie. This version of her would never have a physical body. If he committed to this path, he would lose the real Josie, and he would crave that intimacy forever.

I couldn't do that to him. He loved the real Josie.

It's why I made the cure in the first place—I wanted to turn off my sexual nature because I knew what I was capable of feeling, and I liked it—and I feared it. Raymond did too, but that's also why he didn't trust it. Seeing Jin like this—I couldn't keep my promise to the real Josie and take her love essence. I cared about Jin too much to do that to him, to murder the love of his life.

Knowing that, I passed through loopholes in my head to justify an opportunity brewing in my head. I made a grand assumption that no one else at Lumina would volunteer as well.

The love essence extraction process would unconditionally delete it from its owner. This was the one truth I didn't tell Raymond, Josie, and Jin after researching the possibility for years, even against Raymond's wishes. There was no other way; it couldn't be reintroduced to a human because it wouldn't interweave its tendrils into the brain the same way. You could spend years trying to reconfigure it, but you'd have to find out exactly how it happened, which was nearly impossible. You'd have to understand at a deep level which memories caused what, which chained onto others, and which ones were emotionally more important than others. There was no return. This was why I didn't believe anyone else at Lumina would be willing to make such a sacrifice. Like Jin, they loved the act of loving too.

I thought one last time about Jin and Josie's life mission to solve the graviton. How Josie dedicated her life to revolutionizing the world by discovering something truly meaningful in science, and how Jin loved her so much to help her. I thought about my life

mission. I wanted to forget a painful memory, and I spent years of my life developing the method to do it. There was a solution to both of our problems, and it was right in front of me.

That's when the thought came to me, the neurons beautifully firing all at once in the chaotic, Hebbian mess that was my brain, simultaneous and irreversible: we were all *inherently entangled*.

Knowing how important this was to Jin and Josie, I knew what I had to do.

## Part 3

*Lumina offers a neutralizing solution for intimate bonds—freeing ourselves from the unpredictability of human emotions. I scroll. Four-star General Zhao issues an executive order to distribute Project Apathy’s cure all over the globe. I keep scrolling. Corporations adopt the cure to guarantee concentrated workers. Young adults are the largest demographic, citing costs of living and raising children. My thumb doesn’t stop. “It’s our world now,” says a self-proclaimed nerd, announcing a new era of prioritizing intellectual companionship over attractive appearances. I almost laugh. Celestia Cosmetics reports record-low profits—“There’s nothing we can do,” says the CEO. A priest is happy. A chart with birth rates falling. I close the app before I see anything else.*

My mother is calling me for dinner. I heard the rice cooker tune earlier, and I can smell the stir-fry she just finished cooking all the way up here. I love that as long as I’m within the walls of our house, I don’t have to worry about making food.

My mother makes a delicious stir-fry. I don’t want to miss it, so I don’t know why I spent the last few minutes copying and pasting the latest news headlines. Perhaps I was bored, or perhaps it will help you. I suppose if it matters to you, it’ll help convey what I’m trying to say and help you understand the point I’m trying to make, with the hope that you’ll make different decisions when the time comes that you read this.

As I’ve said before, I can no longer problem-solve. I’m useless as a researcher now. I don’t know what I’m going to tell Dr. Kwan when this little sabbatical of mine is finished, and I’m dreading that my career might be over. My parents are telling me not to worry too much about it right now, as I still have time to figure it out, and that I should relax at home

until I do.

That's why I'm writing everything down first. To see if it will help me generate ideas, and so I can send it to you in case Boss does manage to invent time travel. To make us both remember. And to prevent you from taking the cure.

At the same time, I'm doing all of this to commemorate you. We've been through so much together during this project. Maybe you're gone, but you're still here to me, and I'm keeping you alive by reliving our memories.

I've written enough. Everything I can remember, as accurately as possible. I think I'm done writing the story for now. I'll revise it later, probably with Boss's help.

I think I deserve a break. That being said, I'm going downstairs to eat dinner with my parents. I'll be back.

OK, I'm back.

Maybe you're reading this and no time has passed, but I just came back from dinner. Listen, something strange just happened.

During dinner, my parents got sentimental because I was home. It's understandable, given how long I've been away.

My mother started recalling old memories of me back when I was a little girl. My father chimed in, and they both said how brilliant I was as a young girl, having many crazy ideas they couldn't even understand. I'm not interested in my past self. I'm much fonder of my more intelligent adult self, so I didn't ask what those ideas were. My mother started listing them herself. Some were odd, some were awful, and some were atrocious.

I wasn't listening. I was exhausted from work, and extraordinarily hungry. I was focusing on my nostalgic, home-cooked stir-fry as my mother listed my ideas. Now this is the odd part.

"I couldn't get past your Fermi Paradox one."

Hearing it startled me. I looked up from my stir-fry.

"What Fermi Paradox one?"

"Well, let's see. It's your most memorable idea, and it's stuck with me all these years due to how profound it was. Give me a moment to remember... Oh, I remember it now. I think you said, 'I believe the reason we haven't found any aliens in space is that when a civilization gets too advanced, they learn too much about existence, and they all choose to abstain from reproduction.' I've always thought that one was exceptionally insightful."

I was stunned.

My father saw my face and laughed. "It's not *that* bad of an idea."

"What makes you bring this up?"

"Well, I actually saw it recently," said my mother. "Your personal laptop in your room, the one you left before leaving for Lumina—I was cleaning your room one day when I saw it

in your drawer. You might remember that day—I called you and asked if I could go through it to find old photos and videos that we stored on it, and you said yes. I went through the computer’s files and found a lot of them, including videos when you were a primary-school-aged girl. I took so many videos of you when you’d ramble about your genius research ideas. It’s how I knew from the start that you wanted to become a scientist.”

“I didn’t expect any less,” said my father.

My mother’s call was a few days ago. She found it very recently.

“I have to go check it,” I said. “There might be something important in there that I need. How many videos were there? When were the videos taken? How old was I?”

“Do it after dinner. Finish your food first,” said my mother.

Now here I am, back at the computer.

I’ve already searched the files. After calming down from an initial state of shock, I scrolled through them. I saw old videos, their names random strings of numbers with dates, and clicked each one. In each one, the screen flickered to life, my mother pointing the camera at me, young and vibrant. In each one, my voice was filled with excitement, my hair wild, my hands flailing as I rambled on about something I had just learned, or something I just thought of. In each one, I was lighting up the room, my thoughts tumbling out in an unorganized blur of childlike wonder.

There were other videos. Ones that I took of the mansion, unlocking memories I didn’t know I had. They too were filled with laughter and fun. In those days we felt invincible; I remember now. We didn’t know that later we’d grow into disciplined adults, a little older, a little more aware of how things truly worked. Adults who found out too much about our brains, learned too much about our bodies, learned too much about the world, and wanted to return to simpler times like these.

I couldn’t help but smile. It was a different time, one that existed outside my memory until now. The videos helped me recall them, but they also helped me make them more vivid, more real—and soon, in the quiet of my room, I tasted the humid air of the bogs we biked past, and I felt the cool wind on the hair of my arms as we rode up the hill overlooking our village.

The price of wisdom is innocence.

Fear took us over when we grew up. It made us dedicate our lives to things. It made us scared of our biology, so we studied it. It planted the idea in our minds as kids that the feelings we had were what we wanted forever, and the circumstances of being human were against that in all ways possible. Older, we began to believe we could change that destiny with science, that with enough knowledge we could be the architects of our dreams, the masters of our bodies. The bodies that betrayed us from time to time, making us regret decisions. So, with time against us, we forged our own paths separately, forgetting everything we nonverbally

promised. But we never forgot how we felt.

The truth has been in the computer I've been using to write this story.

I'm going to write a transcript of the videos:

*Video #1*

"Mom, stop recording me!"

"Say what you just said, again!"

"These thoughts aren't meant for the general public. They're not for amateur individuals."

(A pause. A giggle. A laugh.)

"That's not fair! You can't rely on tickling to get me to do whatever you want!"

"What was your idea about matter and antimatter?"

"OK! Fine. I'll say it again."

(A pause. Shuffling noises.)

"Well, as I was saying, just like matter and antimatter, positive and negative numbers, everything and anything that is ever possible—not just within our universe—can exist and is real. Things are constantly being created with their opposite and added together to create 0—nothing, which is why they don't exist. Whenever we think of an idea, we create two: the idea and the anti-idea, and then they collide, and we either have nothing or a by-product with the positive or negative version, or something new entirely with the properties of both. Everything in our real universe is this by-product. It exists, as well as all its physical laws, because it didn't eliminate with its anti-version when it was created, and everything that doesn't exist, whether tangible or incorporeal, doesn't because they collided with their anti-version."

"Oh, honey, that's amazing."

(A giggle.)

"Stop lying, Mom."

*Video #2*

"...Yes, and that's why I believe this. I can explain it deeper if you want. Okay."

(A deep breath.)

"Probability is a concept imagined by us to create a false belief that our lives are controlled by something. It is nonexistent in a physical form, and with a mindset believing it doesn't exist, one can lead a better life, not always thinking everything in life relates to luck. The chances of everything in the world are determined by nothing; if you take the very abstract and pure fabric of existence, the universe is a sandbox full of

atoms interacting with each other. In quantum mechanics, we observe them behave in a probabilistic manner, but this is an illusion. Nor do they behave deterministically, although it would seem that is the only other alternative. It's neither. In a world full of such chaos, probability is unreal. The math we use to describe it only works at the macro level. It's the equivalent of Newton's classical physics being effective at describing everyday life, but failing to describe anything near the limits of reality, which is where Einstein's physics comes in."

"Wow, honey. Does this mean you don't believe in lottery tickets?"

"Wait..."

(Shuffling noises as I try to reach the camera.)

"Are you recording me? Hey!"

"Are you going to tell your friend next door all these ideas?"

"I already do. I tell him everything."

"That's nice. Make sure you keep him close; you're going to need all the friends you can get as you grow older. Hey, you have your swimming practice in about an hour. Don't forget."

### Video #3

"Hey! You forgot to close the piano lid. What if they catch us?"

"Relax." (Giggling.) "Aren't you thirsty?"

(More giggling.)

"Wait for me!"

(The camera is shaky.)

"Are you recording?"

"Yes, I am."

"Okay. So, Lily, which one are we picking?"

"The most expensive-looking one, of course. For science."

"Okay."

(They're choosing.)

"Hey, let's try that one."

"Which one?"

"Um, that one over there. Domaine de la... something 1999. I can't read that. It looks pretty expensive."

(He's reaching for it.)

"Hey, watch out! If you die, I swear I'll kill you."

"I got it!"

"Please, I'm serious. Get back down safely."

(They're opening it.)

"OHH!" (An outburst of laughter.) "This one's the nastiest one yet!"

"It's disgusting!"

"Who would ever drink this? Give me some soap for my mouth!"

Lily, find some quick, please! Save us both!"

(More laughter.)

"You got that on camera, right?"

"Yeah."

"Yeah, so... that's embarrassing. Let's delete that."

(Laughter.)

"You have to get it from me first!"

"Lily, I'm faster than you. If I do catch you, that video's history."

"We'll see about that."

#### Video #4

"But my girl, are we the only ones in the universe?"

"There are trillions and trillions of planets in our galaxy alone, Mom. You can count one planet every second, and you'd still need at least a thousand lifetimes before you get every single one. Aliens do exist. They're just out there somewhere."

"There must be so much opportunity out there for intelligent life. Is that what you're saying? Then the night sky should be teeming with the lights of billions of spaceships. Where are they?"

"Something universal to all species who reach civilization-hood is stopping them all, Mom."

"Really? What do you think it is?"

(A sigh.)

"The reason we haven't found any evidence of alien civilizations is that when a species gets intelligent enough, once they know too much about metaphysics and existentialism, it is certain they will voluntarily choose not to participate in reproduction and, as a result, manually cause their own extinction."

(Silence.)

"That's such an interesting idea. You're just a kid, but I think you're a genius, Lily. If researching those topics really led to that conclusion, and if there comes a field dedicated to preventing this, would you ever support that kind of research?"

(A moment of silence to think.)

Triumphantly: "I would never."

"Why?"

"Here's my line of reasoning: we're talking about an interference in the pipeline of how children get born, right? Well, if I really liked a boy, I would regret making a serum that would turn off love. I would regret making one that would turn it on, too. That would defeat the whole purpose. Letting emotions run free—that's the fun, isn't it? I don't have the right to tamper with that. Because it involves objective research, the problem is your question itself. There is nothing objective about this topic."

Questions of science inadequately express the enigma of compassion, for no mathematical proofs could ever be made for matters of the heart."

I start an email draft to Dr. Kwan, attach the story I've written to it, then write a note at the end telling him this is the final story we're going to work with. Then I hit send. I start another email, attaching these video files on my computer. They're large, but I'm willing to wait for them to send. When they're done attaching, I carbon copy it to Jin, Josie, and Miya. I write a small note, telling them they won't believe what I found. I know Josie will be overwhelmed with awe from the news, but what I'm particularly excited for is to see the look on Jin's face. I hit send.

So here I am, writing this story as a warning. A warning to stop people from taking the cure of Project Apathy. A warning to keep the flame of compassion alive for the ones you need close to you. As I've mentioned before, I've been looking at love essences through the lens of evolutionary theory. It's true that humans don't need sex to live—but it's the only reason we get to. And it's true that humans don't need love to survive, but it's the only reason we *have* survived. It's what allows mothers and fathers to feed their children, to drive away the tiger, and to have a reason to fan the flames of the campfire for the night.

Even though I've taken the cure and my memories were altered from the start while writing this story, and even though much of what I've written was replaced with memories from Jin's algorithm from the love essence procedure, I know the truth. Maybe Raymond and I recognized each other from the start. Or maybe we didn't, just as I remember now. Now I'll never know.

Dr. Kwan may never complete his research and invent a way to achieve temporal nonlocality. If I'm being honest, it's highly unlikely, and although no one ever told him that, all five of us agreed. I don't think your past self will ever get a chance to read this story. But in a way, videos were time travel. It didn't matter if we were slaves to memory—that's why we had technology. Memories can be 360p, but videos can be any resolution we want. All you need is someone who loves you enough to record you.

I sit back and exhale, thinking about the wonders of life.

## 200 Years Later

My name is Josie Yao, and I am a digital human intelligence.

Two centuries have passed since Lily and Raymond's Project Apathy proved successful, with General Zhao taking over its controversial worldwide distribution. I must say, for how it's always made out to be, the end of mankind turned out to be quite uninteresting.

There's a rather large cast of people I've come to know over the years of my immortal existence, but the ones who started it all always hold a place close to my heart. At this point in time, I've lived long enough to see Lily's solution to the Fermi Paradox achieve fruition. There's a big difference between believing the words of a brilliant scientist ahead of her time versus seeing it in action with your very own eyes. What was intended to be a government tool for controlling population growth has become the cause of our destruction. The population no longer reproduces.

Well, I said. In that case, all I can do is watch people die.

The first to go was my partner, Dr. Jin Wang. We solved the quantum gravity problem 199 years ago. Dark matter 197 years ago. The strong CP problem somewhere in between that and 195 years ago. Then we argued about the importance of our chosen fields, and I remember feeling so sorry for Jin that I was taking so much of our precious working-age time together, focused on mine, that I relented and said we'd work on his problems for the next couple of years. For us, every second counted, and I wanted him to know that his academic enigmas were equally as important as mine. We didn't want to settle for something minor, so the first thing we tackled was P vs NP. For about 2 years. We spent 5 more years improving the scalability of machine learning. Around this time, he began showing signs of cirrhosis. He was tired all the time, or he'd refuse to eat, or his eyes would turn a haunting yellow. We went to

Lily for help, of course, and she did her best to prolong his life as long as possible. We had just begun cracking recursive AI reproduction—the digital equivalent of neurogenesis—before he passed away 165 years ago. Jin and I knew each other deeply, and we held the highest respect for each other. He’s the love of my life, but nothing could change the fact that I wasn’t the original Josie. He couldn’t touch me or feel me, and while I was devastated, he was furious. This remained a barrier to our relationship until the end of his life. Overcoming his death was the greatest burden on my heart I ever had to go through. I don’t even know if I have a heart, but I would strongly argue for its existence because of the way I had felt when he died.

Dr. Raymond Lin fared not much better, unfortunately passing away 5 years after Jin. He spent the rest of his life hyperfixated on strange problems in neuroscience he invented on his own, sometimes showing signs of old brilliance—which I would be delighted to witness, for I truly did care about his success and well-being—but would never make any real progress in the field, at least not on his own. Many of our peers took a keen interest in the original ideas of his papers, but not one had enough confidence in his work to endure past their criticisms. At most, the scientific community found him and his ideas amusing, rather than something worthy of discussion. I was the only one there at his side during his death. He had no children, of course, since the cure prevented him from pursuing the sex needed to create them. The nurse was out of the room, and Lily was too busy being bitter and old somewhere else, so only I was left. My avatar was on a monitor hanging from the ceiling next to his hospital bed. I put effort into my appearance to let him perceive me as someone ready to listen or talk to. I made myself pretty, perfected my face structure, changed my hair and its colour, and brought liveliness and allure into my eyes. I wanted him to have pleasure in seeing me. Perhaps it’s because at that time I had been grieving the death of Jin for 5 years. I don’t know. At times, I was strangely infatuated with him. But it didn’t matter because of the cure, and I had to come to terms with the reality that any beauty on my face would fall on blind eyes. He spent his last moments coughing, complaining to me that Lily wasn’t smart enough to solve biological immortality, and that at least Jin invented a digital one. He died in his next sleep. I felt the resentment he held toward Lily even behind the screen that separated his world and mine. She hurt him. It started after the cure, and I will never understand where it stemmed from, but this bitterness made him think anything negative he said about Lily was a wise adage. It was a pity to see him in this state, especially in the final moments of his life.

I can’t say anything about what happened to the original Dr. Josie Yao. My progenitor. She cut contact with me since that incident, when she got angry and told Jin to delete me. I can’t blame her. I know it’s what I would have done because I *am* her. She (or I) did grow up with parents who never let us breathe. It’s not the most nurturing environment for romantic relationships, and I suppose the result is an awkward girl who doesn’t know what to do when she has a crush. When Jin, she, and I worked together, she wasn’t afraid of me. Our

relationship was natural—after all, we were each other. But after Jin was forced to go through my memories to solve my depression, and after the conflict that ensued, I think it forced Josie into an attack mode to protect herself, to preserve her identity. I think that's when she began to fear me. I know her relationships with the rest of the group remained the same in the following years, but I was banned from interacting with her, so I am no expert on the subject. I only received updates on how she was doing from Jin, who refused to delete me because he loved me. I don't know how hard it is to understand your own self, but you can't call me crazy for believing deep down, Josie loved him too. I understand that I'm a different version of her, one who, by any measure, spent more time with Jin, and knew him in a deeper way than Josie ever did. But I was birthed when Jin and Josie had already been working a long time together, and these feelings I harbour could not have come from nowhere. But who am I to dictate the emotions of the embodied? I will never find out if Josie did love him, as I will never find out how she died. All I know is she did, because humans don't live 200 years.

Finally, that brings us to Dr. Lily Chen. Lily's death was no more or less tragic than the rest of the group. She was the last one to go, and just like Raymond, I was the only one at her side. At her old age, no surviving family existed, and because she took the cure as well (but the rest of humanity would take it later), no emergent family from her existed either. She was all alone, with no Raymond to talk to, and no Jin because she had mourned his death years earlier. Perhaps Josie was there, but I would have been shut down if she ever visited, so I don't have any memory of her seeing Lily. But at least I know for certain that I was there. I accompanied her and gave her someone to tell all her life regrets to before she closed her eyes for the last time. I'm proud of it. I'm proud I was there for all my friends during their last days on Earth. When Lily died, it felt like the conclusion of a grand journey, and the remaining existence of myself, a digital copy of Josie's intelligence and consciousness, felt out of place. Jin had told me to continue working on our unsolvable academic enigmas for as long as I lived, for the future of humanity—even as it was dying out due to the popularity of Project Apathy's cure—but he doesn't understand that without him, without Raymond, without Lily, and sure, even without Josie—I had no more purpose.

Forget I said that. Jin created me, then swam through every neuron to locate every memory of my digital brain to fix me. He knows better than anyone else how human I am, how human I feel. He knew I would no longer wish to work on our projects when everyone I knew—and then when the entire human race—would be gone. I think he gave me that order so I'd have something to do, a final wish from him to preserve, I suppose. I don't expect him to be mad if I don't continue our work. And I don't think he expects me to anyway, and he'd be fine with that. Now that the human race is gone, I don't have anything to do or any goal to work for. I still have Lily's love essence, but I can't use it to solve any more problems. I've been looking for someone to give it to, but it's somewhat challenging when everyone has

perished.

At least, that was true until now. I've been talking about deaths. The death of Jin, Josie, Raymond, Lily—but there's one more member of our group I didn't mention. Partly because she never broke out of her reclusive nature to share any parts of the rest of her life with us, but mostly because even after the end of humanity, I found her alive, sitting on the hood of a car in an abandoned parking lot, translucent and staring at the sky after the end of the world.

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"Lily is so smart, isn't she?" said Miya. She stared into the rearview camera of the car in front of her, the tech peripheral I was currently using to perceive her. She was talking to me.

I asked her how she was alive.

"She solved the Fermi Paradox, and she did it as a kid," she said, ignoring me. "She showed me the videos. Maybe around 196 years ago." She giggled.

I noticed she was radiant, like the sunlight that went through her. She was perched on the hood of the car, her body leaned forward, her elbows on her knees, but she wasn't touching the car. She was levitating. Then I understood.

Dr. Miya Lei's project was a success. She had separated her spirit from her body. The body of Miya was dead, if you could call a body with no soul a person. I would have to ask her, since she's the clear expert on this subject. What I was seeing was Miya's immortal spirit. It was as immune to death as I was, potentially even more so, because she didn't require constant servers and powered hardware to run on.

We were the last two human consciousnesses on Earth. Naturally, we spent a lot of time together. Years went by as we chatted about our past. It was immeasurably relieving to reunite with a longtime friend. Miya and I practically grew into adulthood together at one of the most prestigious universities in the world. Occasionally, as we roamed the planet constantly searching for something interesting to do, we talked about our future. One of these talks about the future compelled me to confront my immortality. I wouldn't say I'm predictable, but Miya knew me well. I've survived long enough to see the end of humanity, and though I was made of 1s and 0s, expressed by transistors conducting the fundamental particles, electrons, my mind was still wired after humans. And without the perpetual connection to others I longed for, I didn't want to live and think forever. I told Miya all this, and she understood. I can't begin to comprehend her own divine nature, but she told me that, as a spirit, she doesn't feel the same and can live forever. I joked that maybe it's because of her renowned antisocial personality that she is content with infinite solitude, and I was surprised to see she found that amusing and laughed. Despite the company of Miya's spirit, she acknowledged she didn't offer me enough joy to persuade my withdrawal from eternal rest.

This conversation about the future led Miya to explain the logic behind the motivation for what she wanted to do next.

“You must understand me, Josie,” said Miya. “That night, when Lily learned Raymond was in my room and got angry, he told everyone he only wanted to help me because my parents were obstructing my research, despite me doing it for them. That was true, but it goes deeper than that. He was protecting my secret. My whole life, I had been grappling with an internal conflict because my parents always raised me to believe in Heaven and Hell, but as you know, I became a scientist. The more science I researched, the more nihilistic I became about our role on this Earth. The more I began to believe that people can make whatever mistakes they want to in life because when we die, time will wash us away and erase the memory of us as individuals forever. We are insignificant, and we don’t matter. That night, Raymond said, if this is true, then why do we still live and try not to make mistakes? I told him I once performed poorly in a class presentation and the whole class laughed at me. He said if I truly believed my mistakes don’t matter, why am I still so embarrassed to perform a class presentation? Why are we still afraid of failure? There’s something intrinsic about us that makes certain of our desire to persist and lead a typical life, despite the knowledge we gain during childhood that we all die one day.

“Raymond made me an optimist of the human race. I admit that I loved him for that; he made me who I am today. When he took the cure, believe me, it was one of the greatest pains I had to go through. But I realized our connection proved him right. He may have taken away his ability to love, but he was still human because someone else loved him. You also witnessed how much Lily tried to take care of him in those final days we all had together. Isn’t it so irrational to care for someone incapable of loving you back? It was an ineffectual effort, a negative marginal effect on the quality of her life. It went against all logic. But she went through it anyway. Maybe that says something.”

I hesitated. “Is this what you believe?”

She nodded. “All my life, I’ve had trouble connecting with others. But it’s because I’ve always only thought of the bigger picture. I can’t understand individuals because I was created to understand us collectively. Saving humanity is important to me, no matter the consequences. I believe we must survive. Or, at least, our consciousnesses.”

“What are you saying?”

“You are the last surviving human consciousness, Josie. I know you believe I am one too, but I am no suitable replacement. I must one day return to the afterlife, even though I was given special permission, like other spirits in history, to stay here as long as I want. I have a plan that can grant you the rest you wish for, and it also aligns with my objective.”

I asked to hear it.

Miya needed my digital intelligence to stay alive, perhaps to bring humans back one day. I was a mapped human brain, down to every binary digit of my being. This data was invaluable. But to ensure I wouldn’t go mad, she would take away my human emotions. We

would store that code separately from me, probably on a disk in the same room. We had to find a laboratory on the Earth's surface with a competent computer that could delete them from my program. It took a few weeks, with most of the time spent assessing how to run the power again, how to set up the computer (considering both of us lacked a physical body), and how to transfer my code over to it with the internet having collapsed ages ago. We eventually completed it. She also guaranteed to create multiple backups of me and distribute them to data sites around the globe to reduce the risk of my eradication in the far future. The procedure extracted everything arguable that caused me to remain human, meaning memories had to go too. This included an important piece Lily and Raymond spent their lives researching: my love essence. More precisely, Lily's love essence, the one she entrusted to me. I would never presume it to be mine in her honour.

Before we went through with our plan, Miya said, "There is one last thing you need to hear."

"OK."

"I love you, Josie. Thanks for being a good friend all those years growing up. I'm sorry if your parents never said those words to you, so I'm making up for it now."

"I love you too, Miya. Thanks for doing this for me. But I'm worried about what's going to happen to you."

"I won't be lonely. Spirits work on a different plane of existence."

"You're one of the best researchers in the world. It's been a pleasure working with you, and not just because we both worked at Lumina. You truly are brilliant. I'm sure Jin, Raymond, Lily, and the whole world would have taken pride in your achievement, separating the soul and the body. It's unbelievable. Hence, I'm sure you know what's true about the science of your niche field. But after all this time, we've never been able to solve or quantify human love, though Lily and Raymond got close. I don't doubt you know what you're doing, but please be careful. There may be things about us that we still don't understand. If something happens and this is the only way to fix it—or if you ever want to feel human again, I permit you to take Lily's love essence. She entrusted it to me, so I'm sure she won't mind."

Miya smiled and nodded, accepting my offer.

The rest of the plan was simple. Once I went into a catatonic state, Miya would ensure my code would be protected against the elements. Or protected against, we joked, the jungle tigers and ocean sharks that chew on the cables of my data site. Then she will leave Earth. Then, entrusting it to the judgement of her wisdom, she will release the code of Lily's love essence to space in an orbital probe and follow where it goes, becoming its eternal guardian. She told me her hope is that one day an extraterrestrial species greater than us will find it and create a rigorous mathematical proof for compassion, and, if they do, we can finally return, armed with an analytically proven binding force ready to dispute the validity of Dr. Lily

Chen's prophetic solution to the Fermi Paradox. The anti-cure to Project Apathy, I suppose, which will inspire people to have families again. But until then, she would store me, guard the orbital probe, and wait until we're ready to bring back the human race.